dirac matrices deflating — geometry, Dirac matrices, spin connection, Dirac equations and their solutions on the 4+4 and 7+1 spacetimes whose coordinates x5, x6, x7 deflate exponentially

PROJECT spacetimes, Revision_Deflation. Provenance: definitions, derivations, machine checks, and how to reproduce them

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0. What this document proves, and for whom

The main project studies a 16-component Dirac field on two flat eight-dimensional spacetimes: one of signature **4+4** (four space-like and four time-like coordinates) and one of signature **7+1** (seven space-like coordinates and one time-like coordinate). Its document `dirac matrices` (in the folder above this one) displays the eight real 16×16 Dirac matrices of each spacetime, proves their properties and writes down the Dirac equations of the fields **dirac1644** and **dirac1671**.

The revision replaces the two flat spacetimes by two curved ones, in which three of the eight coordinates shrink as the ordinary time advances. This document repeats, for the curved spacetimes, everything that `dirac matrices` does for the flat ones. It is written for a reader who has worked through `dirac matrices` and who knows general relativity at the level of a first course: metric, Christoffel symbols, curvature, geodesics. What such a course usually omits — how a spinor is differentiated on a curved spacetime — is derived here from the beginning (Part 2).

- **Part 1** is geometry: the metric, its Christoffel symbols, its curvature, its Einstein tensor, its symmetries and its geodesics.
- **Part 2** constructs the Dirac matrices of the curved spacetimes and the spin connection, which is the object that tells a spinor how to follow the curvature.
- **Part 3** writes down the Dirac equations of the two fields of the revision, **dirac1644d** and **dirac1671d**, and brings them to the form in which they are solved.
- **Part 4** solves them, mode by mode, in closed form where that is possible and numerically where it is not. Two results stand out. On the 7+1 spacetime the deflation *creates particles*, with a thermal spectrum whose temperature is the rate of deflation divided by 2π. On the 4+4 spacetime the deflation *extinguishes* every dependence of the field on the three deflating times.
- **Part 5** states which symmetries of the flat spacetimes survive.
- **Part 6** separates what has been established from what has not.

How the statements are verified. Each statement of Parts 1 to 3 is an identity between rational functions of the rates and of the scale factors. The Rust program `deflation` (folder `code/deflation-rs`) verifies each identity in exact rational arithmetic at five sets of rational values of the parameters; each verification has an identifier such as `44d-C4` (the prefix `44d-` refers to the deflating 4+4 spacetime, `71d-` to the deflating 7+1 spacetime, `def-` to both). An independent Python script verifies the same identities *symbolically*, for arbitrary values of the parameters, by computer algebra; its identifiers begin with the letter `I`. The statements of Part 4 are verified numerically by two different integration methods in two languages. The complete output of all programs is reproduced

verbatim in Appendix A.

0.1 The hypotheses

The hypotheses are the author's. In his words, the coordinates $x5$, $x6$ and $x7$ *deflate exponentially*, on the 4+4 spacetime and on the 7+1 spacetime alike, and Minkowski space is replaced by a pseudo-Riemannian space with a diagonal metric. For the 4+4 spacetime he specified the three components of the metric that change:

$$g_{55} = -\,\mathrm{const5}^2\, e^{-2\,\mathrm{g55}\,x4}, \qquad g_{66} = -\,\mathrm{const6}^2\, e^{-2\,\mathrm{g66}\,x4}, \qquad g_{77} = -\,\mathrm{const7}^2\, e^{-2\,\mathrm{g77}\,x4}, \tag{0.1}$$

with three positive rates g55, g66, g77 and three positive constants, which may differ from one another. (The author writes one word, const, for the three constants; because they may differ they are given three names here.) The other five components of the metric keep their flat values.

The names. g55, g66 and g77 are the author's names for the three **rates**. They are written upright and without a subscript throughout the revision. The three **components of the metric** are written with subscripts, g_{55}, g_{66}, g_{77}. Equation (0.1) shows the two side by side: the rate g55 is the number in the exponent of the component g_{55} of the metric.

The 7+1 spacetime. For the 7+1 spacetime the author wrote the same three exponentials with a positive sign, which is their form in a convention that gives space-like components of the metric the positive sign. The main project writes the 7+1 metric with the opposite convention, $\eta_{1671} = \mathrm{diag}(-1,-1,-1,-1,+1,-1,-1,-1)$, in which the time-like component is positive. In that convention the three components of the metric are again those of Eq. (0.1), with the minus signs. The geometry is the same in either convention: the proper length of a coordinate interval along $x5$ is $\mathrm{const5}\, e^{-\mathrm{g55}\,x4}$ times the coordinate interval. Equation (0.1) therefore holds on both spacetimes in the conventions of this project; what differs is the meaning of the sign. On the 7+1 spacetime $x5$, $x6$, $x7$ are three of the seven space-like coordinates, and what deflates is a *length*. On the 4+4 spacetime they are three of the four time-like coordinates, and what deflates is a *duration* measured along an additional time.

What "deflates" means. The coordinate $x4$ is the ordinary time. As $x4$ increases by $1/\mathrm{g55}$, every proper interval along $x5$ shrinks by the factor e. The word is the opposite of inflation, in which lengths grow exponentially.

Tags. The flat spacetimes and their objects carry the tags 1644 and 1671 in the main project. The deflating spacetimes and their objects carry the tags **1644d** and **1671d**. A general coordinate among the three deflating ones is written xt with $t \in \{5,6,7\}$, and its rate is written rate_t, so that $\mathrm{rate}_5 = \mathrm{g55}$, $\mathrm{rate}_6 = \mathrm{g66}$, $\mathrm{rate}_7 = \mathrm{g77}$. A general coordinate among $x0, x1, x2, x3$ is written xa.

0.2 The unique symbols of this document

The main project keeps a MASTER registry of symbols in which every symbol has one meaning. The revision adds the symbols below to it; the provenance RP1 proves by machine that none of them collides with a symbol of the main registry or with a symbol of the author's notebooks. The column "in programs and data files" gives the spelling of the same object in the Rust program and in the data files.

symbol	in programs and data files	meaning	defined in	origin
g55, g66, g77	`g55`, `g66`, `g77`	the three rates of deflation of $x5$, $x6$, $x7$ (positive constants, dimension 1/length). These are the author's names; they are written upright and never with a subscript. They are *not* components of the metric: the components are g_{55}, g_{66}, g_{77}	RP0 1; DMD 0.1; M I	author
rate_t	`rate_t`, `rate`	the rate of the coordinate xt in a statement that holds for $t = 5, 6, 7$ alike: $\mathrm{rate}_5 = \mathrm{g55}$, $\mathrm{rate}_6 = \mathrm{g66}$, $\mathrm{rate}_7 = \mathrm{g77}$	DMD 0.1; M I	new

symbol	in programs and data files	meaning	defined in	origin
gsum	`gsum`, `gam`	g55 + g66 + g77, the total rate: the proper volume of the three deflating coordinates falls as $e^{-\mathrm{gsum}\,x4}$ (`gam`: gsum in units of 100 km/s/Mpc)	DMD 1.1; M II	new
gsq, gprod	`gsq`, `gprod`	$\mathrm{g55}^2 + \mathrm{g66}^2 + \mathrm{g77}^2$ and $\mathrm{g55\,g66} + \mathrm{g55\,g77} + \mathrm{g66\,g77}$; $\mathrm{gsum}^2 = \mathrm{gsq} + 2\,\mathrm{gprod}$	DMD 1.3; M II	new
const5, const6, const7; const_t	`const5`, `const6`, `const7`, `const_t`, `cst`	the three positive constants of the author's metric (his word: const): the values of the scale factors at $x4 = 0$	RP0 1; DMD 0.1; M I	author
b_5, b_6, b_7; b_t	`b_5`, `b_6`, `b_7`, `b_t`	the scale factors of the deflating coordinates, $b_t(x4) = \mathrm{const}_t\, e^{-\mathrm{rate}_t\, x4}$: a coordinate distance dxt is the proper length $b_t\,\lvert dxt\rvert$. Always written with the index 5, 6, 7 or t (the letter b without index is a Ginzburg-Landau coefficient of the main registry)	DMD 1.1; M II	new
V_d	`Vd`	$b_5 b_6 b_7 = \sqrt{\lvert\det g\rvert}$, the proper volume of a unit coordinate cell of the deflating coordinates; $V_d \propto e^{-\mathrm{gsum}\,x4}$	DMD 1.1; RS1 1; M II	new
g_{1644d}, g_{1671d}; $g_{\mu\nu}$	`g_1644d`, `g_1671d`	the metrics of the deflating 4+4 and 7+1 spacetimes, $\mathrm{diag}(\eta_{00}, \dots, \eta_{44}, \eta_{55}b_5^2, \eta_{66}b_6^2, \eta_{77}b_7^2)$ with $\eta = \eta_{1644}$ or η_{1671}; in both, $g_{tt} = -\mathrm{const}_t^2 e^{-2\,\mathrm{rate}_t\, x4}$	DMD 1.1; M II	author
μ, ν, ρ, σ (indices)	`mu`, `nu`, `rho`, `sigma`, `lam`	coordinate indices with the values $0, \dots, 7$ on the deflating spacetimes (only as sub- or superscripts; as stand-alone letters μ, ν, λ keep the meanings of the main registry). The capital indices A, B, C, D of the main registry, also with a prime, label the orthonormal frame: the frame vector A points along the coordinate xA (as a stand-alone letter C is the coupling of the main registry)	DMD 1.1; M II	new
t, s (indices)	—	indices of the three deflating coordinates, values 5, 6, 7: t as in the main registry, and s as a second such index where two are needed (s_h with the index h are matrices of the main registry)	DMD 0.1, 1.3; M I, II	main
a, b; j (indices)	—	a, b as in the main registry, with the range stated at each place: in RS1 to RS4 and in the manuscripts the index of the four extra axes, values 0, 5, 6, 7 (on the deflating spacetimes a frame index); in DMD and in the statements of the checks of the programs the index of the four flat coordinates other than the time, values 0, 1, 2, 3. In the manuscripts of the revision, where both sets of coordinates occur, the flat coordinates other than the time carry the index j with the range $0, 1, 2, 3$ stated there; a sum over j with other limits (the ordinary space $1, 2, 3$ of the main registry; in DMD Eq. (3.12) the seven space-like directions) states them	DMD 0.1; RS1 5.1; M I	main
${e_A}^{\mu}$, ${e^A}_{\mu}$	`e_A^mu`	the orthonormal frame and its inverse: ${e^A}_{\mu} = \mathrm{diag}(1, 1, 1, 1, 1, b_5, b_6, b_7)$	DMD 1.1; M II	new
$\Gamma^{\rho}{}_{\mu\nu}$	`Gamma`, `christoffel`	the Christoffel symbols of the deflating metric (three indices; the letter Γ with square brackets is the effective action of the main registry)	DMD 1.2; M II	new
$R^{\rho}{}_{\sigma\mu\nu}$, $R^{\mu}{}_{\nu}$, R	`riemann`, `ricci`, `Ric44`, `Ric55`, `Ric66`, `Ric77`, `Rscalar`	Riemann tensor, Ricci tensor and curvature scalar of the deflating metric	DMD 1.3; M II	new

symbol	in programs and data files	meaning	defined in	origin
$\mathcal{G}^{\mu}{}_{\nu}$	`Ein00`, `Ein44`, `Ein55`, `Ein66`, `Ein77`, `einstein_mixed`	the Einstein tensor $R^{\mu}{}_{\nu} - \frac{1}{2} R\, \delta^{\mu}_{\nu}$ (script letter, so that it cannot be taken for Newton's constant or for a constant of the Fermi type)	DMD 1.4; M II	new
$\mathcal{X}$; $\mathcal{X}_{\text{scale}}$, $\mathcal{X}_{(s)}$	`killing_vectors`, `killing_count`	a Killing vector field of the deflating metric (script letter; the plain letter X is the set of coordinates of the main registry); the scaling vector $\partial_4 + \sum_t \text{rate}_t\, xt\, \partial_t$; the three special Killing vectors of the case of equal rates, $s = 5, 6, 7$	DMD 1.5, 1.6; M II, App. B	new
$\bar{x}4$; $\bar{x}5$, $\bar{x}6$, $\bar{x}7$	—	coordinates of the case of equal rates in which the line element of $x4, \ldots, x7$ is a multiple of a flat one: $\bar{x}4 = e^{\text{g55}\, x4}/\text{g55}$, $\bar{x}t = \text{const}_t\, xt$	DMD 1.6; M App. B	new
$\mathcal{H}^{\mu}{}_{\nu}$; α_{GB}, Λ_8	`Lanczos`, `alpha_GB`, `Lambda_8`	the Lanczos tensor of the Gauss-Bonnet term (script letter; H is the Hubble rate); the coupling of the Gauss-Bonnet term and the cosmological constant of Einstein-Gauss-Bonnet gravity in eight dimensions (with the index 8; Λ without index is the momentum cutoff of the main registry)	DMD 1.4a; M II	new
K_{AB}	`ksec`	the sectional curvature of the plane of the frame directions A and B, in the convention in which the space-like components of the metric are positive: $K_{4t} = \text{rate}_t^2$, $K_{ts} = \text{rate}_t\, \text{rate}_s$ for $t \neq s$, zero if one direction is flat. Always with two subscripts: the letter K without subscript is the generator of the main registry, and K with an order as subscript and with an argument is a modified Bessel function	DMD 1.4a; M App. B	new
ς	`affine`	affine parameter along a geodesic; for a massive particle on the 7+1 spacetime, the proper time divided by the mass	DMD 1.7; M II	new
P_{μ}	`P_1`, `P_5`	the conserved covariant momenta of a geodesic (with an index; P without index is the generator of the main registry)	DMD 1.7; M II	new
$x4_{\text{turn}}$	`x4_turn`	the value of the ordinary time at which a classical particle with momentum along a deflating time turns back, and beyond which a mode that depends on a deflating time is extinguished (4+4): there $k_d = E_{\perp}$	DMD 1.7, 4.3; M II (4+4)	new

symbol	in programs and data files	meaning	defined in	origin
γ^{μ}_{1644d}, γ^{μ}_{1671d}	`gamma_1644d`, `gamma_1671d`	the Dirac matrices of the deflating spacetimes in coordinate components, $\gamma^{\mu} = e_A{}^{\mu} \gamma^A$: $\gamma^t = \gamma^{A=t}/b_t$ for $t = 5, 6, 7$, the other five are the constant matrices γ^A_{1644}, γ^A_{1671} of the main project	DMD 2.2; M III	new
$\omega_{\mu AB}$	`omega_(mu, A, B)`	the coefficients of the spin connection (three indices; ω_{1644}, ω_{1671} with a tag are the volume elements of the main registry)	DMD 2.3; M III	new
Ω_{μ}	`Omega_mu`, `omega`	the spin connection as a matrix, $\frac{1}{4} \omega_{\mu AB} \gamma^A \gamma^B$; the covariant derivative of a spinor is $\nabla_{\mu} \psi = (\partial_{\mu} + \Omega_{\mu}) \psi$	DMD 2.3; M III	new
ψ_{1644d}, ψ_{1671d}	`dirac1644d`, `dirac1671d`	the fields dirac1644d and dirac1671d: the 16-component fields of the main project on the deflating spacetimes; $(\gamma^{\mu} \nabla_{\mu} - m) \psi_{1644d} = 0$ and $(i \gamma^{\mu} \nabla_{\mu} - m) \psi_{1671d} = 0$	DMD 3.1, 3.2; M III	new

symbol	in programs and data files	meaning	defined in	origin
$\hat{\psi}$	psihat	the weighted field $\sqrt{V_d}\,\psi$, which obeys the equation without spin connection	DMD 3.3; M III	new
$K(x4)$, $\mathsf{h}(x4)$	K, h	the generator K_{1644} and the Hamiltonian matrix h_{1671} of the main registry with the wavenumbers along the deflating coordinates divided by the scale factors: $\partial_4\hat{\psi} = K\hat{\psi}$ (4+4), $i\partial_4\hat{\psi} = \mathsf{h}\hat{\psi}$ (7+1)	DMD 3.4; M III	main

symbol	in programs and data files	meaning	defined in	origin
k_5, k_6, k_7; k_t	k_5, k_6, k_7, k_t	the conserved wavenumbers of a mode along the deflating coordinates (covariant components of the wave vector k_A of the main project)	DMD 3.4; M III	new
$E_\perp$	E_perp	the energy of a mode, or of a classical particle, without its motion along the deflating coordinates: $\sqrt{m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2}$ (with the wavenumbers or momenta along $x0, \ldots, x3$)	DMD 1.7, 3.4; M II, III	new
k_d	k_d	the physical wavenumber along the deflating coordinates, $k_d(x4) = \left[\sum_t k_t^2/b_t(x4)^2\right]^{1/2}$; it grows as the coordinates deflate	DMD 3.4; M III	new
$\mathsf{h}_\perp$, $K_\perp$	h_perp, K_perp	the parts of h and of K that do not contain the scale factors: the operators of the modes that do not depend on the deflating coordinates, $\mathsf{h}_\perp = m\beta + \sum_a p_a\alpha^a$ with $\mathsf{h}_\perp^2 = E_\perp^2$, and $K_\perp = -m\gamma^4 + i\sum_a k_a\gamma^4\gamma^a$ with $K_\perp^2 = -E_\perp^2$	DMD 4.1; M III	new
γ^d, α^d	—	the frame matrix along the direction of the wave vector in the deflating coordinates, $\gamma^d = k_d^{-1}\sum_t (k_t/b_t)\gamma^t$, and $\alpha^d = \gamma^4\gamma^d$; constant in time when one rate governs	DMD 4.1; M III	new
w; f_+, f_-; Z_+, Z_-	—	auxiliary objects of the exact solutions (DMD 4.2 and 4.3; w, Z_+, Z_- also in Sec. IV of the manuscripts): a constant eigenvector of $\mathsf{h}_\perp$ or of $K_\perp$; the two coefficient functions of the ansatz; two cylinder functions whose orders differ by one	DMD 4.2, 4.3; M IV	new
u_d	u_d	k_d/rate_t for a mode that depends on one deflating coordinate: the argument of the Bessel functions of the exact solutions	DMD 4.1; M IV	new
n_{cr}	n_cr, n_min, n_mean, n_max, n_exact, fd5, fd6	the number of particles per internal state that the deflation creates in a mode that starts empty (7+1)	DMD 4.2; M IV (7+1)	new
T_{defl}	—	the temperature of the thermal spectrum of the created quanta, $\mathrm{rate}_t/2\pi$ in units with $\hbar = k_B = 1$ (7+1)	DMD 4.2; M IV (7+1)	new
S_{WKB}	S_wkb	$\int \sqrt{k_d^2 - E_\perp^2}\,dx4$ from the turning value on: exponent of the extinction of a bounded solution (4+4)	DMD 4.3; M IV (4+4)	new
φ_{bos}, $\hat{\varphi}_{\mathrm{bos}}$, m_{bos}	phi_bos, Shat, Shat2, m_bos	a mode of the Vorton-Scalar (the radial mode, $m_{\mathrm{bos}} = m_s$, or the phase mode, $m_{\mathrm{bos}} = 0$), its weighted form $\sqrt{V_d}\,\varphi_{\mathrm{bos}}$, and its mass	RS1 2; M V	new
ϖ_b	varpi_b	$\sqrt{k^2 + m_{\mathrm{bos}}^2 - \mathrm{gsum}^2/4}$: the frequency of the weighted boson mode that does not depend on the deflating coordinates	RS1 2; M V	new

Symbols that this document takes over unchanged from the main project — the matrices γ^A_{1644} and

γ^A_{1671}, the flat metrics η_{1644} and η_{1671}, the matrices C^- (`Cm1644`), ω_{1671}, β, α^A, τ^A, $\bar\tau^A$ and the masses — are defined in `dirac matrices` and in the provenance P1 of the main project.

0.3 How to reproduce everything in this document

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
# 1. build the Rust program of the revision and run its unit tests
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
cargo test  --release --manifest-path code/deflation-rs/Cargo.toml
# 2. geometry, matrices, spin connection, Dirac operators; solutions; symmetries
./code/deflation-rs/target/release/deflation matrices
./code/deflation-rs/target/release/deflation modes
./code/deflation-rs/target/release/deflation symmetry
# 3. independent verification (computer algebra and general-purpose integrators)
python3 code/scripts/verify_deflating_independent.py
# 4. the four figures of this document
python3 code/scripts/make_figures.py geodesics isometries creation extinction
# 5. rebuild this document (Markdown, LaTeX, PDF)
python3 code/scripts/assemble_docs.py "docs_src/dirac matrices deflating.src.md"
bash code/scripts/build_docs.sh "dirac matrices deflating.md"
```

Step 2 writes the files listed in Appendix B. Each command prints its checks and ends with a line of the form "47 of 47 checks passed"; it returns a non-zero exit status if a check fails. The provenance RP8 explains every step for a reader who has not used a terminal before.

Part 1. Geometry of the deflating spacetimes

1.1 Metric, orthonormal frame, volume

Let η stand for $\eta_{1644} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ or for $\eta_{1671} = \mathrm{diag}(-1,-1,-1,-1,+1,-1,-1,-1)$, the flat metrics of the main project, with the coordinates in the order $x0,\dots,x7$. Define the three **scale factors**

$$b_5(x4) = \mathrm{const5}\, e^{-\mathrm{g55}\, x4}, \qquad b_6(x4) = \mathrm{const6}\, e^{-\mathrm{g66}\, x4}, \qquad b_7(x4) = \mathrm{const7}\, e^{-\mathrm{g77}\, x4}. \tag{1.1}$$

The metric of the deflating spacetime is

$$g_{\mu\nu} = \mathrm{diag}\big(\eta_{00},\, \eta_{11},\, \eta_{22},\, \eta_{33},\, \eta_{44},\, \eta_{55}\, b_5^2,\, \eta_{66}\, b_6^2,\, \eta_{77}\, b_7^2\big), \tag{1.2}$$

called g_{1644d} when $\eta = \eta_{1644}$ and g_{1671d} when $\eta = \eta_{1671}$. Because $\eta_{55} = \eta_{66} = \eta_{77} = -1$ in both cases, the three components $g_{tt} = -b_t^2$ of the metric are those of Eq. (0.1). They obey $\partial_4 g_{tt} = -2\,\mathrm{rate}_t\, g_{tt}$ (checks `44d-G1`, `71d-G1`).

Greek indices μ,ν,ρ,σ label the coordinates and take the values $0,\dots,7$. Capital indices A, B label an **orthonormal frame**: eight vector fields $e_A = e_A{}^\mu \partial_\mu$ that are orthogonal and of unit length at every point, $g_{\mu\nu} e_A{}^\mu e_B{}^\nu = \eta_{AB}$. For a diagonal metric the simplest frame points along the coordinates:

$$e_A{}^\mu = \mathrm{diag}\Big(1,1,1,1,1,\frac{1}{b_5},\frac{1}{b_6},\frac{1}{b_7}\Big), \qquad e^A{}_\mu = \mathrm{diag}(1,1,1,1,1,b_5,b_6,b_7). \tag{1.3}$$

The second matrix is the inverse of the first; its rows are the eight one-forms $e^A = e^A{}_\mu dx^\mu = (dx0,\dots,dx4,\ b_5\,dx5,\ b_6\,dx6,\ b_7\,dx7)$. The frame vector e_t is $b_t^{-1}\partial_t$: one unit of proper length (or of proper duration) along xt is the coordinate interval $1/b_t$, which *grows* as the coordinate deflates.

The volume element is $\sqrt{|\det g|}\; d^8x = V_d\, d^8x$ with

$$V_d(x4) = b_5\, b_6\, b_7 = \mathrm{const5}\,\mathrm{const6}\,\mathrm{const7}\; e^{-\mathrm{gsum}\, x4}, \qquad \mathrm{gsum} = \mathrm{g55} + \mathrm{g66} + \mathrm{g77}. \tag{1.4}$$

V_d is the proper volume of a unit coordinate cell of the three deflating coordinates. The total rate gsum is the rate at which this volume shrinks; it is the single combination of the three rates that enters most results of the revision.

1.2 Christoffel symbols

The Christoffel symbols are $\Gamma^{\rho}{}_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$. The only components of the metric that depend on a coordinate are the three g_{tt}, and they depend on $x4$ alone. A Christoffel symbol can therefore be different from zero only if its three indices are $t, t, 4$ in some order, with the same t. Two cases remain. First, $\rho = t$:

$$\Gamma^{t}{}_{4t} = \Gamma^{t}{}_{t4} = \tfrac{1}{2}g^{tt}\partial_4 g_{tt} = -\mathrm{rate}_t. \tag{1.5}$$

Second, $\rho = 4$, where only the last term of the definition contributes:

$$\Gamma^{4}{}_{tt} = -\tfrac{1}{2}g^{44}\partial_4 g_{tt} = \eta_{44}\,\mathrm{rate}_t\, g_{tt} = \eta_{44}\eta_{tt}\,\mathrm{rate}_t\, b_t^2. \tag{1.6}$$

There is no sum over t in these formulas. The product $\eta_{44}\eta_{tt}$ is -1 on the 7+1 spacetime and $+1$ on the 4+4 spacetime; this sign is the origin of every difference between the two spacetimes in what follows. All other Christoffel symbols vanish (checks `44d-G2`, `71d-G2`; symbolically `I44d-G1`, `I71d-G1`).

1.3 Curvature

Method. The curvature is obtained most quickly from the frame, by Cartan's structure equations. The connection one-forms $\omega^A{}_B$ are defined by the two conditions $de^A + \omega^A{}_B \wedge e^B = 0$ (no torsion) and $\omega_{AB} = -\omega_{BA}$ (the frame stays orthonormal), where $\omega_{AB} = \eta_{AC}\,\omega^C{}_B$. From Eq. (1.3), $de^t = d(b_t)\wedge dxt = -\mathrm{rate}_t\, e^4 \wedge e^t$, and $de^A = 0$ for the other five one-forms. The two conditions are met by

$$\omega^{t}{}_4 = -\mathrm{rate}_t\, e^t, \qquad \omega^{4}{}_t = \eta_{44}\eta_{tt}\,\mathrm{rate}_t\, e^t \qquad (t = 5, 6, 7), \tag{1.7}$$

with all other connection one-forms zero. Indeed $de^t + \omega^t{}_4 \wedge e^4 = -\mathrm{rate}_t\, e^4\wedge e^t - \mathrm{rate}_t\, e^t\wedge e^4 = 0$, and $de^4 + \sum_t \omega^4{}_t \wedge e^t = 0$ because $e^t \wedge e^t = 0$. The curvature two-forms are $R^A{}_B = d\omega^A{}_B + \omega^A{}_C \wedge \omega^C{}_B = \frac{1}{2}R^A{}_{BCD}\, e^C \wedge e^D$. Two kinds are different from zero. For $t \neq s$,

$$R^t{}_4 = -\mathrm{rate}_t\, de^t = \mathrm{rate}_t^2\; e^4 \wedge e^t, \qquad R^t{}_s = \omega^t{}_4 \wedge \omega^4{}_s = -\eta_{44}\eta_{ss}\,\mathrm{rate}_t\,\mathrm{rate}_s\; e^t \wedge e^s.$$

Reading off the components in the orthonormal frame:

$$R^t{}_{4t4} = -\mathrm{rate}_t^2, \qquad R^4{}_{t4t} = -\eta_{44}\eta_{tt}\,\mathrm{rate}_t^2, \qquad R^t{}_{sts} = -\eta_{44}\eta_{ss}\,\mathrm{rate}_t\,\mathrm{rate}_s \quad (t \neq s), \tag{1.8}$$

together with the components that follow from the antisymmetry in the last two indices. No component carries an index 0, 1, 2 or 3: the four coordinates $x0, x1, x2, x3$ span a flat space, and the curvature lives entirely in the four coordinates $x4, x5, x6, x7$.

The program computes the Riemann tensor twice, once from the Christoffel symbols and once from Eq. (1.7), and verifies that the two results agree (checks `44d-G3`, `71d-G3`) and that both equal Eq. (1.8) (checks `44d-G4`, `71d-G4`; symbolically `I44d-G3`, `I71d-G3`).

Ricci tensor and curvature scalar. Two more combinations of the rates are needed:

$$\mathrm{gsq} = \mathrm{g55}^2 + \mathrm{g66}^2 + \mathrm{g77}^2, \qquad \mathrm{gprod} = \mathrm{g55}\,\mathrm{g66} + \mathrm{g55}\,\mathrm{g77} + \mathrm{g66}\,\mathrm{g77},$$

which are related to the total rate by $\mathrm{gsum}^2 = \mathrm{gsq} + 2\,\mathrm{gprod}$. Contracting Eq. (1.8), $R_{BD} = R^A{}_{BAD}$, and raising one index with η:

$$\begin{aligned} R^4{}_4 &= -\eta_{44}\,\mathrm{gsq}, \qquad R^t{}_t = -\eta_{44}\,\mathrm{rate}_t\,\mathrm{gsum}, \qquad R^a{}_a = 0 \;\; (a = 0, 1, 2, 3), \\ R &= -\eta_{44}(\mathrm{gsq} + \mathrm{gsum}^2). \end{aligned} \tag{1.9}$$

Because the frame points along the coordinates, the mixed components $R^{\mu}{}_{\nu}$ in the coordinate basis are the same numbers. The curvature scalar is constant: it depends neither on the place nor on the time.

1.4 The Einstein tensor: the deflating spacetimes are not vacuum solutions

The Einstein tensor is $\mathcal{G}^{\mu}{}_{\nu} = R^{\mu}{}_{\nu} - \frac{1}{2} R\, \delta^{\mu}_{\nu}$. From Eq. (1.9),

$$\mathcal{G}^{4}{}_{4} = \eta_{44}\, \mathrm{gprod}, \qquad \mathcal{G}^{t}{}_{t} = \eta_{44}\Big[\tfrac{1}{2}(\mathrm{gsq} + \mathrm{gsum}^2) - \mathrm{rate}_t\, \mathrm{gsum}\Big], \qquad \mathcal{G}^{a}{}_{a} = \tfrac{1}{2}\,\eta_{44}(\mathrm{gsq} + \mathrm{gsum}^2), \tag{1.10}$$

and its covariant divergence vanishes, as the Bianchi identity requires (checks `44d-G5`, `71d-G5`; symbolically `I44d-G2`, `I71d-G2`). For three equal rates, $\mathrm{g55} = \mathrm{g66} = \mathrm{g77}$, the three expressions are $3\,\mathrm{g55}^2$, $3\,\mathrm{g55}^2$ and $6\,\mathrm{g55}^2$ times η_{44}.

Two conclusions follow.

1. *The Einstein tensor does not vanish.* The deflating spacetime is not a solution of Einstein's equations in vacuum.
2. *A cosmological constant does not help.* A cosmological constant alone would require $\mathcal{G}^{\mu}{}_{\nu}$ to be a multiple of δ^{μ}_{ν}. Equating $\mathcal{G}^{4}{}_{4}$ with $\mathcal{G}^{a}{}_{a}$ gives $2\,\mathrm{gprod} = \mathrm{gsq} + \mathrm{gsum}^2$, that is $\mathrm{gsq} = 0$: all three rates would vanish.

If gravity in eight dimensions obeys *Einstein's* equations, the deflation therefore requires a source with the stress tensor $\mathcal{G}^{\mu}{}_{\nu}/(8\pi G_8)$, where G_8 is the eight-dimensional constant of gravitation: a positive energy density proportional to gprod on the 7+1 spacetime, and a stress along the flat coordinates $x0, \ldots, x3$ that is larger than the energy density. The hypotheses prescribe the metric and say nothing about its source. This document follows them: the metric (1.2) is a *given background*, on which the fields of the project move without acting back on it. The next subsection shows that in the gravity of Lovelock, which the author's notebooks study, no source is needed if the three rates are equal.

1.4a What can sustain the deflation: the Gauss–Bonnet term

In more than four dimensions Einstein's equations are not the only field equations of second order for the metric. The general ones are those of Lovelock, which add to the Einstein tensor a series of tensors of higher order in the curvature; the first of them is the Lanczos tensor $\mathcal{H}^{\mu}{}_{\nu}$ of the Gauss–Bonnet term. The author's notebooks on the Einstein–Lovelock equations (N06 to N08 of the provenance P3 of the main project) study this theory. It is therefore natural to ask whether the deflating spacetime solves the field equations of Einstein–Gauss–Bonnet gravity without matter,

$$\mathcal{G}^{\mu}{}_{\nu} + \Lambda_8\, \delta^{\mu}_{\nu} + \alpha_{GB}\, \mathcal{H}^{\mu}{}_{\nu} = 0, \tag{1.10a}$$

which follow from the action $R - 2\Lambda_8 + \alpha_{GB}\,(R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})$ with a cosmological constant Λ_8 and a coupling α_{GB}. (In this subsection, and only here, the signs are those of the convention in which the space-like components of the metric are positive: the metric g_{1644d} itself, and minus the metric g_{1671d}. In that convention the Einstein tensor is $-\eta_{44}$ times that of Eq. (1.10), for both spacetimes.)

The Lanczos tensor. Computed from its definition,

$$\mathcal{H}^{a}{}_{a} = -4\,\mathrm{g55}\,\mathrm{g66}\,\mathrm{g77}\,\mathrm{gsum} \quad (a = 0, 1, 2, 3), \qquad \mathcal{H}^{4}{}_{4} = \mathcal{H}^{t}{}_{t} = 0, \tag{1.10b}$$

for both spacetimes, and the Lovelock tensor of the third order, the last one that can be different from zero in eight dimensions, vanishes identically (checks `44d-G10`, `71d-G10`, by a formula for curvatures that act diagonally on pairs of directions; independently `I44d-L1`, `I71d-L1`, from the standard formula of the Lanczos tensor by computer algebra).

The formula that the program uses, so that the result can be checked by hand. In the orthonormal frame the Riemann tensor (1.8) has, for each pair of frame directions $A \neq B$, one independent

component: the sectional curvature $K_{AB} = K_{BA}$ of the plane that the two directions span. In the convention of this subsection

$$K_{4t} = \mathrm{rate}_t^2, \qquad K_{ts} = \mathrm{rate}_t\,\mathrm{rate}_s \quad (t \neq s), \tag{1.10c}$$

and $K_{AB} = 0$ for every pair that contains one of the flat directions $0, 1, 2, 3$. (A sectional curvature carries two subscripts; the letter K without subscript is the matrix of Eq. (3.9).) For a Riemann tensor of this kind the Einstein tensor and the Lanczos tensor are diagonal, and

$$\mathcal{G}^A{}_A = -\sum_{\{B,C\}} K_{BC}, \qquad \mathcal{H}^A{}_A = -4 \sum_{\{B,C\},\{B',C'\}} K_{BC}\,K_{B'C'} \tag{1.10d}$$

without summation over A. The first sum runs over the pairs of directions that do not contain the direction A; the second runs over the unordered couples of two such pairs that have no direction in common, so that a set of four directions contributes its three divisions into two pairs. *Test:* for a space of constant curvature in eight dimensions, where all K_{BC} equal one number, the first sum has 21 terms and the second 3×35, and the formulas give -21 times that number and -420 times its square: the known values $-\frac{1}{2}\cdot 7\cdot 6$ and $-\frac{1}{2}\cdot 7\cdot 6\cdot 5\cdot 4$. *Application:* a couple of pairs contributes to $\mathcal{H}^A{}_A$ only if neither pair contains a flat direction, that is, only if the four directions are $4, 5, 6, 7$. For a flat direction A these four are allowed, and their three divisions give

$$K_{45}K_{67} + K_{46}K_{57} + K_{47}K_{56} = \mathrm{g55}\,\mathrm{g66}\,\mathrm{g77}\,\mathrm{gsum},$$

which is Eq. (1.10b). For $A = 4, 5, 6, 7$ the direction A is itself one of the four, no couple remains, and the component vanishes. The Lovelock tensor of the third order is built in the same way from sets of six directions other than A; every such set contains a flat direction, and the tensor vanishes. With Eq. (1.10c) the first formula gives $\mathcal{G}^4{}_4 = -\mathrm{gprod}$ and $\mathcal{G}^a{}_a = -(\mathrm{gsq} + \mathrm{gprod})$, which are Eq. (1.10) multiplied by $-\eta_{44}$, as they must be.

The result. Insert Eqs. (1.10) and (1.10b) into Eq. (1.10a).

- The components along $x4$ and along the three deflating coordinates contain no Gauss–Bonnet term. They require the four components $\mathcal{G}^4{}_4, \mathcal{G}^5{}_5, \mathcal{G}^6{}_6, \mathcal{G}^7{}_7$ of the Einstein tensor to be equal. Because $\mathcal{G}^5{}_5 - \mathcal{G}^6{}_6 = \pm(\mathrm{g55} - \mathrm{g66})\,\mathrm{gsum}$, this holds **if and only if the three rates are equal**; the cosmological constant is then $\Lambda_8 = 3\,\mathrm{g55}^2$.
- The components along the flat coordinates then fix the coupling: $\alpha_{GB} = -1/(4\,\mathrm{g55}^2)$.

Hence:

> **With three equal rates, the deflating 4+4 spacetime and the deflating 7+1 spacetime are exact solutions of Einstein–Gauss–Bonnet gravity without matter, with $\Lambda_8 = 3\,\mathrm{g55}^2$ and $\alpha_{GB} = -1/(4\,\mathrm{g55}^2)$, that is, $4\,\alpha_{GB}\Lambda_8 = -3$.** With rates that are not all equal they are not, for any Λ_8 and α_{GB}.

Meaning. In the theory that the author's notebooks study, the deflation at equal rates needs no source: the rate is fixed by the constants of the theory, $\mathrm{g55} = \sqrt{\Lambda_8/3}$, and the theory must have the particular ratio $4\alpha_{GB}\Lambda_8 = -3$ of its two constants. Solutions of Einstein–Gauss–Bonnet gravity in which every scale factor is an exponential of the time have been classified in five and six dimensions (Ref. [1]); the present solution is of that kind in eight dimensions, with four static and three contracting directions. Three limits of the result must be stated. (i) It concerns the metric alone; the fields of Parts 3 and 4 remain test fields on it. (ii) Whether the solution is stable against perturbations of the metric has not been examined. (iii) If gravity is of this kind, the relation between Newton's constant and the volume of the deflating coordinates that the supplement RS3 assumes for Einstein gravity (its assumption A2) would have to be derived anew.

1.5 Symmetries of the metric

A Killing vector is a vector field $\mathcal{X} = \mathcal{X}^\rho \partial_\rho$ along which the metric does not change: $\mathcal{X}^\rho \partial_\rho g_{\mu\nu} + g_{\rho\nu}\partial_\mu \mathcal{X}^\rho + g_{\mu\rho}\partial_\nu \mathcal{X}^\rho = 0$. Flat eight-dimensional spacetime has 36 of them (8 translations and 28 rotations and boosts). The deflating spacetime has fewer.

For any rates there are 14 Killing vectors.

- *Seven translations*, along every coordinate except $x4$: the metric does not depend on these coordinates.
- *Six rotations* of the four flat coordinates, $xa\,\partial_b - xb\,\partial_a$ with $a < b$ in $\{0,1,2,3\}$.
- *One scaling*,

$$\mathcal{X}_{\text{scale}} = \partial_4 + \mathsf{g55}\,x5\,\partial_5 + \mathsf{g66}\,x6\,\partial_6 + \mathsf{g77}\,x7\,\partial_7. \tag{1.11}$$

  It generates the transformation that shifts the ordinary time $x4$ by an arbitrary constant and multiplies each deflating coordinate xt by the exponential of rate_t times that constant. The shift of the time shrinks every proper interval along xt, and the stretching of the coordinate xt undoes it.

Equal rates add more. If two rates are equal, $\text{rate}_t = \text{rate}_s$, the rotation of the two coordinates that share the rate, $(\text{const}_t/\text{const}_s)\,xt\,\partial_s - (\text{const}_s/\text{const}_t)\,xs\,\partial_t$, is a fifteenth Killing vector. If all three rates are equal there are 20: the 14, the three rotations of $x5, x6, x7$, and three further vectors that Sec. 1.6 constructs (checks `44d-G6`, `71d-G6`; symbolically `I44d-G4`, `I71d-G4`).

The lists are complete. A Killing vector is determined by its value and its first derivatives at a single point. The values span at most 8 dimensions. A Killing vector that vanishes at the point is determined by its first derivatives there, which form a matrix of the Lorentz algebra of the tangent space; and this matrix must leave the Riemann tensor at the point unchanged. With one index raised, the Riemann tensor (1.8) acts on antisymmetric pairs of frame vectors as a diagonal matrix: $e_t \wedge e_4$ has the eigenvalue $-\eta_{44}\,\text{rate}_t^2$, $e_t \wedge e_s$ has the eigenvalue $-\eta_{44}\,\text{rate}_t\,\text{rate}_s$, and every pair that contains a flat direction e_a has the eigenvalue zero. The matrix must map each eigenspace to itself. The eigenspace of the eigenvalue zero singles out the flat directions, so the matrix is the sum of a rotation of $(e_0, \dots, e_3)$ — six parameters, always allowed — and a transformation of $(e_4, \dots, e_7)$. The latter moves $e_4 \wedge e_t$ into combinations of $e_s \wedge e_t$ and $e_4 \wedge e_s$, whose eigenvalues differ from that of $e_4 \wedge e_t$ unless rates coincide. For three different positive rates no such transformation is allowed; for exactly two equal rates only the rotation of the two equal directions is; for three equal rates all six are. The number of Killing vectors is therefore at most $8 + 6 = 14$, $8 + 7 = 15$ and $8 + 12 = 20$, which are the numbers exhibited (checks `I44d-G5`, `I71d-G5`, by exact computation of ranks).

Three consequences.

1. *The deflating spacetime is homogeneous.* The seven translations and the scaling carry any point to any other point. No place and no moment is distinguished; in particular, the curvature is the same everywhere.
2. *The constants are not observable by themselves.* A change of const_t is undone by a rescaling of the coordinate xt, and so is a shift of $x4$. The deflation becomes observable only through an object that fixes a coordinate scale along xt: a wave with a given coordinate wavelength (Part 4), or a coordinate period if xt is periodic.
3. *There is no conserved energy.* A time translation ∂_4 alone is not a symmetry. What is conserved along the scaling (1.11) is the combination $P_4 + \sum_t \text{rate}_t\,xt\,P_t$ of the momenta of Sec. 1.7. Particle numbers and energies can and do change with time (Part 4).

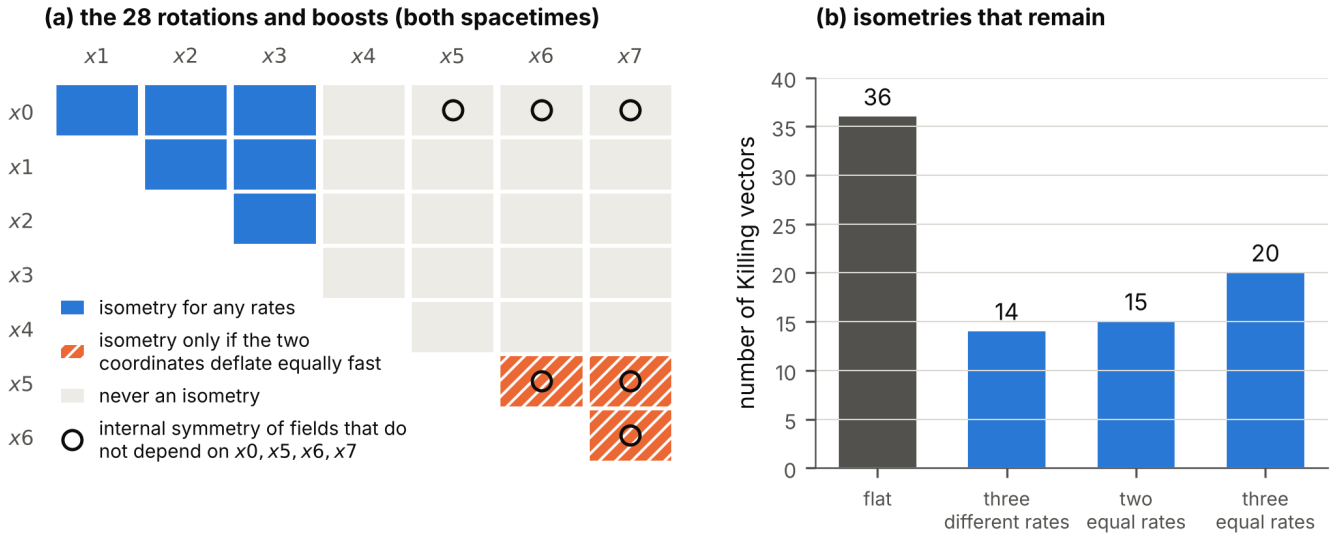


Figure 1: The symmetries of the flat and of the deflating spacetimes. (a) The 28 rotations and boosts of the flat spacetime, one square for each pair of coordinates: those that remain isometries for any rates (the six rotations of $x0, x1, x2, x3$), those that remain isometries only if the two coordinates deflate equally fast, and those that are never isometries of the deflating spacetime. A circle marks the six pairs whose rotations act as internal symmetries on fields that do not depend on $x0, x5, x6, x7$ (Part 5). The diagram is the same for 4+4 and for 7+1. (b) Number of Killing vectors: 36 on the flat spacetime; 14, 15 and 20 on the deflating spacetime with three different, two equal and three equal rates. Data: `data/isometries.csv`; program: `deflation symmetry`.

1.6 Equal rates: spaces of constant curvature

Let $\mathrm{g55} = \mathrm{g66} = \mathrm{g77}$. Introduce four new coordinates in place of $x4, x5, x6, x7$:

$$\bar{x}4 = \frac{e^{\mathrm{g55}\,x4}}{\mathrm{g55}}, \qquad \bar{x}t = \mathrm{const}_t\, xt \quad (t = 5, 6, 7),$$

so that $dx4 = d\bar{x}4/(\mathrm{g55}\,\bar{x}4)$. The part of the line element that contains $x4, \ldots, x7$ becomes

$$\eta_{44}\, dx4^2 - e^{-2\,\mathrm{g55}\,x4} \sum_t d\bar{x}t^2 = \frac{\eta_{44}\, d\bar{x}4^2 - \sum_t d\bar{x}t^2}{(\mathrm{g55}\,\bar{x}4)^2}. \tag{1.12}$$

- **7+1** ($\eta_{44} = +1$): the line element (1.12) is that of de Sitter space of radius $1/\mathrm{g55}$, in the half of its flat slicing in which space *contracts*. De Sitter space is familiar as the geometry of inflation, where the same line element is used with the opposite direction of time. The deflating 7+1 spacetime with equal rates is the product of flat four-dimensional space $(x0, x1, x2, x3)$ with a contracting four-dimensional de Sitter space.
- **4+4** ($\eta_{44} = -1$): the line element (1.12) is *minus* that of four-dimensional hyperbolic space H^4 of radius $1/\mathrm{g55}$, in the upper-half-space form. The four time-like coordinates span a negative-definite space of constant curvature.

In both cases the Riemann tensor of the four coordinates has the form of a space of constant curvature, $R_{\rho\sigma\mu\nu} = -\eta_{44}\,\mathrm{g55}^2\,(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu})$ (checks `44d-G7`, `71d-G7`).

The three special Killing vectors. The line element (1.12) is a multiple of the flat one, $\eta_{44}\, d\bar{x}4^2 - \sum_t d\bar{x}t^2$. The special conformal transformations of that flat metric along the three axes $\bar{x}5, \bar{x}6, \bar{x}7$ leave the factor $1/\bar{x}4^2$ unchanged, and are therefore isometries. Their generators are, for $s =$

$5, 6, 7,$

$$\mathcal{X}_{(s)} = 2\,\bar{x}s\,\Big[\bar{x}4\,\frac{\partial}{\partial \bar{x}4} + \sum_t \bar{x}t\,\frac{\partial}{\partial \bar{x}t}\Big] - \Big(\sum_t \bar{x}t^2 + \eta_{44}\eta_{tt}\,\bar{x}4^2\Big)\frac{\partial}{\partial \bar{x}s}, \qquad \bar{x}4\,\frac{\partial}{\partial \bar{x}4} = \frac{1}{\mathsf{g55}}\,\partial_4. \quad (1.13)$$

The sign $\eta_{44}\eta_{tt}$ is $+1$ for 4+4 and -1 for 7+1.

1.7 Geodesics: what a classical particle does

The translations give seven conserved momenta $P_\mu = g_{\mu\nu}\,dx^\nu/d\varsigma$, $\mu \neq 4$, where ς is an affine parameter along the path (for a massive particle on the 7+1 spacetime, the proper time divided by the mass). The norm of the momentum is also conserved, and gives the velocity along $x4$. Write $E_\perp = \sqrt{m^2 + \sum_a P_a^2}$ for the energy that the particle would have without momentum along the deflating coordinates (the same quantity appears for waves in Part 3). Then

$$\Big(\frac{dx4}{d\varsigma}\Big)^2 = E_\perp^2 \;\pm\; \sum_t \frac{P_t^2}{b_t(x4)^2}, \qquad +\text{ for 7+1}, \quad -\text{ for 4+4}. \quad (1.14)$$

The term P_t^2/b_t^2 grows as $e^{2\,\mathrm{rate}_t\,x4}$: the conserved momentum P_t belongs to a fixed *coordinate* wavelength, and the proper wavelength shrinks. The two spacetimes now behave in opposite ways.

7+1: unbounded blueshift. The energy $dx4/d\varsigma$ of a particle that moves along a deflating coordinate grows without bound. For motion along $x5$ alone, Eq. (1.14) integrates in closed form: the affine parameter that elapses between $x4$ and $x4 = \infty$ is

$$\varsigma(\infty) - \varsigma(x4) = \frac{1}{\mathsf{g55}\,E_\perp}\,\mathrm{arsinh}\Big(\frac{E_\perp\,\mathrm{const5}\,e^{-\mathsf{g55}\,x4}}{|P_5|}\Big), \quad (1.15)$$

which is finite. A particle with momentum along a deflating coordinate reaches $x4 = \infty$ after a *finite proper time*. The coordinates of Eq. (1.2) do not cover its whole future: the deflating 7+1 spacetime is geodesically incomplete. For equal rates this is the well-known fact that the flat slicing covers only half of de Sitter space. In the example of check `71d-G8` ($m = 1$, $P_1 = 0.5$, $P_5 = 0.3$, $\mathsf{g55} = 1$, $\mathrm{const5} = 0.8$, start at $x4 = -2$) the proper time to $x4 = \infty$ is 3.386358.

4+4: a turning value. With the minus sign, the right-hand side of Eq. (1.14) vanishes where $\sum_t P_t^2/b_t^2 = E_\perp^2$. For motion along $x5$ alone this happens at

$$x4_{\mathrm{turn}} = \frac{1}{\mathsf{g55}}\,\ln\frac{E_\perp\,\mathrm{const5}}{|P_5|}, \quad (1.16)$$

and the affine parameter needed to reach it from $x4$ is $\mathrm{arcosh}\big(E_\perp\,\mathrm{const5}\,e^{-\mathsf{g55}\,x4}/|P_5|\big)/(\mathsf{g55}\,E_\perp)$. The path reaches $x4_{\mathrm{turn}}$ and returns to smaller $x4$. A classical particle with momentum along a deflating *time* cannot pass the turning value of the ordinary time. In the example of check `44d-G9` (same numbers) the turning value is 1.09240103, reached after the affine parameter 3.38543615; the numerical integration reproduces both.

The wave equation of Part 4 shows the same two behaviours in quantum form: creation of particles of ever larger energy on 7+1, and total reflection — the extinction of the dependence on the deflating times — on 4+4.

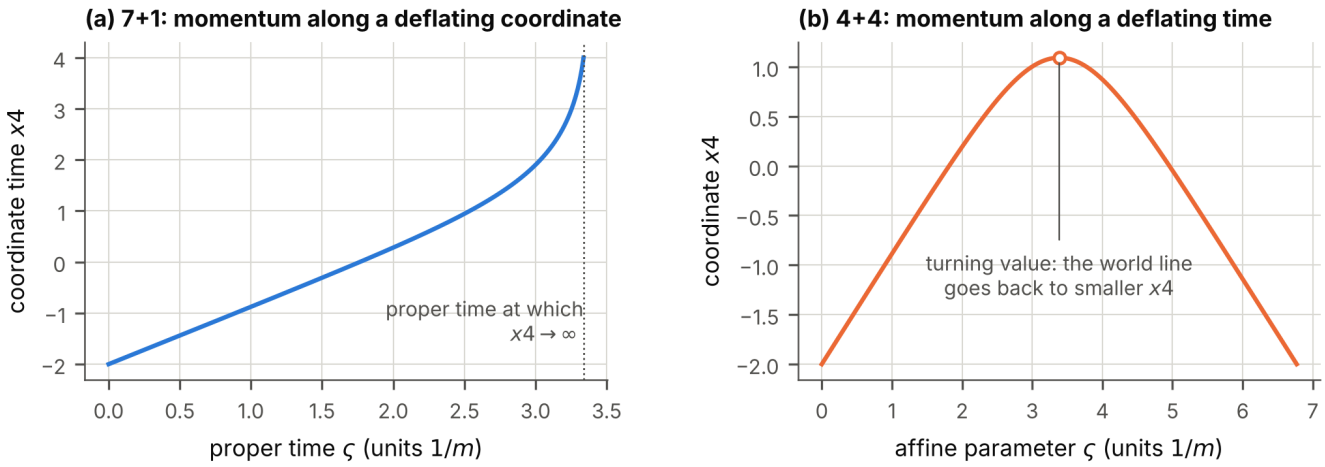


Figure 2: Geodesics with momentum along the deflating coordinate $x5$ (the examples of checks `71d-G8` and `44d-G9`). (a) 7+1: the ordinary time $x4$ against the proper time; the path runs to $x4 = \infty$ within the finite proper time given by Eq. (1.15). (b) 4+4: $x4$ against the affine parameter; the path turns at the value of Eq. (1.16). Data: `data/geodesic_1671d.csv`, `data/geodesic_1644d.csv`; program: `deflation matrices`.

Part 2. Dirac matrices and spin connection

2.1 The matrices of the frame are the matrices of the main project

At each point of a curved spacetime the tangent space is a flat space with the metric η_{AB}. The Dirac matrices of the orthonormal frame are therefore the constant matrices of the flat spacetime: the eight real 16×16 matrices γ^A_{1644} on the deflating 4+4 spacetime and γ^A_{1671} on the deflating 7+1 spacetime, with

$$\gamma^A\gamma^B + \gamma^B\gamma^A = 2\,\eta^{AB}\,1.$$

Nothing about them changes. The program of the revision nevertheless executes all 106 checks of the main project on them again, so that the revision does not rest on a result that it has not itself verified (first part of the command `deflation matrices`; Appendix A.4).

2.2 The matrices in coordinate components

The Dirac matrices with a coordinate index are

$$\gamma^\mu = e_A{}^\mu\,\gamma^A: \qquad \gamma^\mu = \gamma^{A=\mu}\ \ (\mu = 0,\dots,4), \qquad \gamma^t = \frac{1}{b_t(x4)}\,\gamma^{A=t}\ \ (t = 5,6,7). \tag{2.1}$$

They are called γ^μ_{1644d} and γ^μ_{1671d}. Five of them are the constant matrices of the main project; the three along the deflating coordinates are the constant matrices divided by the scale factors, and so *grow* with $x4$. They are real, and they satisfy the Clifford relations with the inverse of the curved metric,

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\,g^{\mu\nu}\,1, \qquad g^{tt} = \frac{\eta_{tt}}{b_t^2}, \tag{2.2}$$

for all 64 pairs of indices (checks `44d-C1`, `71d-C1`; symbolically `I44d-C1`, `I71d-C1`). To avoid ambiguity, in every formula below in which a scale factor appears explicitly the matrices γ^4, γ^a, γ^t are the *constant* matrices of the frame.

2.3 The spin connection

Why it is needed. A spinor is defined relative to a frame. On a curved spacetime the frame at one point cannot be compared with the frame at a neighbouring point without a rule, and the derivative of a spinor must contain a term that accounts for the turning of the frame from point to point. That term is the spin connection.

Its coefficients. The components of the connection one-forms of Sec. 1.3, $\omega^A{}_B = \omega_\mu{}^A{}_B\, dx^\mu$, are given in terms of the Christoffel symbols by the "tetrad postulate"

$$\omega_\mu{}^A{}_B = e^A{}_\nu(\partial_\mu e_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}\, e_B{}^\lambda). \tag{2.3}$$

With Eqs. (1.3), (1.5) and (1.6), and with the first index lowered, $\omega_{\mu AB} = \eta_{AA}\,\omega_\mu{}^A{}_B$, the only coefficients different from zero are

$$\omega_{t\,t4} = -\omega_{t\,4t} = -\eta_{tt}\,\mathrm{rate}_t\, b_t \qquad (t = 5, 6, 7), \tag{2.4}$$

where the first index is the coordinate index and the other two are frame indices. For example, $\omega_t{}^t{}_4 = e^t{}_t\,\Gamma^t{}_{t4}\,e_4{}^4 = -\mathrm{rate}_t\, b_t$. The coefficient with $\mu = 4$ vanishes because the two terms of Eq. (2.3) cancel: $\partial_4(1/b_t) = +\mathrm{rate}_t/b_t$ and $\Gamma^t{}_{4t}/b_t = -\mathrm{rate}_t/b_t$ (checks `44d-C2`, `71d-C2`; Eq. (2.4) agrees with Eq. (1.7)).

As matrices. The spin connection acts on spinors through the matrices

$$\Omega_\mu = \tfrac{1}{4}\,\omega_{\mu AB}\,\gamma^A\gamma^B: \qquad \Omega_t = -\tfrac{1}{2}\,\eta_{tt}\,\mathrm{rate}_t\, b_t\,\gamma^t\gamma^4 \quad (t = 5, 6, 7), \qquad \Omega_0 = \cdots = \Omega_4 = 0, \tag{2.5}$$

(checks `44d-C3`, `71d-C3`; symbolically `I44d-C2`, `I71d-C2`). The covariant derivative of a spinor is

$$\nabla_\mu\psi = (\partial_\mu + \Omega_\mu)\,\psi.$$

Each Ω_t is a multiple of the generator $\gamma^t\gamma^4$ of the Lorentz transformation in the plane of xt and $x4$. On 7+1 this is a boost, on 4+4 a rotation of two time-like directions.

The test of correctness. The spin connection is right if and only if the Dirac matrices are covariantly constant, that is, if differentiating the vector index with the Christoffel symbols and the two spinor indices with Ω_μ gives zero:

$$\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\,\gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0 \qquad \text{for all 64 pairs } (\mu, \nu), \tag{2.6}$$

with the matrices γ^ν of Eq. (2.1). This is verified (checks `44d-C4`, `71d-C4`; symbolically `I44d-C3`, `I71d-C3`). Property (2.6) is what makes the current of the Dirac field conserved and the Dirac equation covariant.

The one combination that enters the Dirac equation. Contracting (2.5) with the matrices (2.1) and using $(\gamma^t)^2 = \eta_{tt}$ for the frame matrices,

$$\sum_\mu \gamma^\mu\,\Omega_\mu = -\tfrac{1}{2}\sum_t \eta_{tt}\,\mathrm{rate}_t\,(\gamma^t)^2\gamma^4 = -\tfrac{1}{2}\,\mathrm{gsum}\;\gamma^4 \tag{2.7}$$

(checks `44d-C5`, `71d-C5`; symbolically `I44d-C4`, `I71d-C4`). The three scale factors have cancelled, and only the total rate remains.

2.4 The curvature of the spinor derivative

Two covariant derivatives of a spinor do not commute; their commutator is the Riemann tensor acting on the spinor:

$$[\nabla_\mu, \nabla_\nu]\,\psi = (\partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu])\psi = \tfrac{1}{4}\,R_{AB\mu\nu}\,\gamma^A\gamma^B\,\psi. \tag{2.8}$$

The program verifies this for all 64 pairs with the Riemann tensor that it computed from the metric (checks `44d-C6`, `71d-C6`). This is a third, independent route to the curvature, through the spinors. Contracting once more,

$$\tfrac{1}{2}\,\gamma^{\mu}\gamma^{\nu}\,[\nabla_{\mu},\nabla_{\nu}] = -\frac{R}{4}\,1, \qquad R = -\eta_{44}(\mathrm{gsq} + \mathrm{gsum}^2) \tag{2.9}$$

(checks `44d-C7`, `71d-C7`; symbolically `I44d-C5`, `I71d-C5`). Consequently the square of the Dirac operator $\gamma^{\mu}\nabla_{\mu}$ is the wave operator of the spinor derivative minus $R/4$: the curvature acts on a spinor like a constant addition to the square of the mass, of magnitude $(\mathrm{gsq} + \mathrm{gsum}^2)/4$.

Part 3. The Dirac equations of the fields dirac1671d and dirac1644d

The Dirac equations of the main project contain the operator $\gamma^{A}\partial_{A}$. On a curved spacetime the partial derivative of a spinor is replaced by the covariant derivative, and the constant matrices by the matrices (2.1). This "minimal" replacement is the only one that keeps the equation covariant and its current conserved without introducing a new constant; it is adopted here.

3.1 The equation of dirac1671d

The field **dirac1671d**, $\psi_{1671d}(x)$, is a column of sixteen complex-valued Grassmann operators on the deflating 7+1 spacetime. Its equation is

$$\boxed{(i\,\gamma^{\mu}_{1671d}\,\nabla_{\mu} - m)\,\psi_{1671d} = 0} \tag{3.1}$$

With Eq. (2.7), and written with the constant matrices of the frame,

$$i\,\gamma^{4}\Big(\partial_{4} - \frac{\mathrm{gsum}}{2}\Big)\psi + i\sum_{a=0}^{3}\gamma^{a}\partial_{a}\psi + i\sum_{t=5}^{7}\frac{1}{b_t(x4)}\,\gamma^{t}\partial_{t}\psi - m\,\psi = 0. \tag{3.2}$$

Compared with the flat equation (2.9) of `dirac matrices`, two things have changed: the derivative along the ordinary time is shifted by $\mathrm{gsum}/2$, and each derivative along a deflating coordinate is divided by its scale factor.

3.2 The equation of dirac1644d

The field **dirac1644d**, $\psi_{1644d}(x)$, is a column of sixteen complex-valued Grassmann operators on the deflating 4+4 spacetime. Its equation has no factor i, for the reason given in Sec. 1.8.1 of `dirac matrices`:

$$\boxed{(\gamma^{\mu}_{1644d}\,\nabla_{\mu} - m)\,\psi_{1644d} = 0} \tag{3.3}$$

that is,

$$\gamma^{4}\Big(\partial_{4} - \frac{\mathrm{gsum}}{2}\Big)\psi + \sum_{a=0}^{3}\gamma^{a}\partial_{a}\psi + \sum_{t=5}^{7}\frac{1}{b_t(x4)}\,\gamma^{t}\partial_{t}\psi - m\,\psi = 0. \tag{3.4}$$

3.3 The weighted field

The shift of ∂_4 is removed by a change of the normalization of the field. Define the **weighted field**

$$\hat{\psi} = \sqrt{V_d}\;\psi. \tag{3.5}$$

Because $\partial_4\sqrt{V_d} = -\frac{1}{2}\,\mathrm{gsum}\,\sqrt{V_d}$ by Eq. (1.4), $\sqrt{V_d}\,(\partial_4 - \mathrm{gsum}/2)\,\psi = \partial_4\hat{\psi}$. The weighted field therefore obeys Eqs. (3.2) and (3.4) *without* the shift:

$$\text{7+1: } \Big(i\gamma^4\partial_4 + i\sum_a \gamma^a\partial_a + i\sum_t \frac{\gamma^t}{b_t}\partial_t - m\Big)\hat{\psi} = 0, \qquad \text{4+4: } \Big(\gamma^4\partial_4 + \sum_a \gamma^a\partial_a + \sum_t \frac{\gamma^t}{b_t}\partial_t - m\Big)\hat{\psi} = 0. \tag{3.6}$$

Meaning. The quantity $\psi^\dagger\psi$ is a number of particles per unit of proper volume, and $\hat{\psi}^\dagger\hat{\psi} = V_d\,\psi^\dagger\psi$ is the number per unit of *coordinate* volume. When the deflating coordinates shrink, the same particles occupy less proper volume and $\psi^\dagger\psi$ grows as $e^{\mathrm{gsum}\,x4}$ for that reason alone. The weighted field removes this trivial growth.

The central observation. *A field that does not depend on $x5$, $x6$, $x7$ does not feel the deflation at all, apart from the weight:* for such a field Eqs. (3.6) are exactly the flat equations of the main project. Every result of the main project for fields of this kind — in particular, everything about the pairing of fermions — remains true for the weighted field. The deflation enters such results in one way only, through the dependence of the *couplings* on V_d, which the supplement RS1 derives. A field that does depend on a deflating coordinate feels the deflation directly, through the factors $1/b_t$; Part 4 treats it.

3.4 Modes, and the operators h and K

The coefficients of Eqs. (3.6) depend on $x4$ only. A solution can therefore be taken proportional to $\exp\big(i\sum_a p_a\,xa + i\sum_t k_t\,xt\big)$, with seven constant wavenumbers: four along the flat coordinates and three, k_5, k_6, k_7, along the deflating coordinates. (The wavenumbers along the flat coordinates are called p_a on 7+1 and k_a on 4+4, as in the main project. The wavenumbers k_t are called κ_t in `dirac matrices`, where they are frequencies along the additional times of the flat 4+4 spacetime.) A solution of this kind is a **mode**. What remains is an ordinary differential equation in $x4$ for sixteen functions.

7+1. Multiplying the first of Eqs. (3.6) by γ^4, with $(\gamma^4)^2 = 1$, $\beta = \gamma^4$ and $\alpha^A = \gamma^4\gamma^A$:

$$i\,\partial_4\hat{\psi} = \mathsf{h}(x4)\,\hat{\psi}, \qquad \mathsf{h}(x4) = m\,\beta + \sum_{a=0}^{3} p_a\,\alpha^a + \sum_{t=5}^{7}\frac{k_t}{b_t(x4)}\,\alpha^t. \tag{3.7}$$

The matrix h is real and symmetric for every $x4$, and

$$\mathsf{h}^2 = (E_\perp^2 + k_d^2)\,1, \qquad E_\perp = \sqrt{m^2 + \textstyle\sum_a p_a^2}, \qquad k_d(x4) = \sqrt{\sum_t \frac{k_t^2}{b_t(x4)^2}} \tag{3.8}$$

(checks `7d-D1`, `7d-D2`; symbolically `I7d-D1`). $E_\perp$ is the energy that the mode would have without the deflating coordinates. k_d is the *physical* wavenumber along the deflating coordinates; it grows with $x4$, like $e^{\mathrm{rate}_t\,x4}$, because the coordinate wavelength $2\pi/k_t$ is fixed while the proper length shrinks. The evolution in $x4$ is unitary, and $\hat{\psi}^\dagger\hat{\psi}$ is conserved; but the Hamiltonian depends on the time, and its derivative $d\mathsf{h}/dx4 = \sum_t \mathrm{rate}_t\,(k_t/b_t)\,\alpha^t$ anticommutes with β (check `7d-D3`). A time-dependent Hamiltonian that is not diagonal in the states of definite energy creates pairs.

4+4. Multiplying the second of Eqs. (3.6) by γ^4, with $(\gamma^4)^2 = -1$:

$$\partial_4\hat{\psi} = K(x4)\,\hat{\psi}, \qquad K(x4) = -m\,\gamma^4 + i\sum_{a=0}^{3} k_a\,\gamma^4\gamma^a + i\sum_{t=5}^{7}\frac{k_t}{b_t(x4)}\,\gamma^4\gamma^t, \tag{3.9}$$

$$K^2 = (k_d^2 - E_\perp^2)\,1, \qquad E_\perp = \sqrt{m^2 + \textstyle\sum_a k_a^2} \tag{3.10}$$

(checks `44d-D1`, `44d-D2`; symbolically `I44d-D1`). The sign in Eq. (3.10) is the opposite of that in Eq. (3.8), and with it the behaviour changes. While $k_d < E_\perp$ the square of K is negative and the mode oscillates. Once the physical wavenumber has grown beyond $E_\perp$ the square of K is positive, and the solutions grow or decay exponentially (the worked example of check `44d-D7`). The first two terms of K are anti-Hermitian and the third is Hermitian; K is anti-Hermitian, and the evolution unitary, if and only if $k_5 = k_6 = k_7 = 0$ (check `44d-D5`).

The weighted field also obeys a second-order equation, $\partial_4^2\hat{\psi} = (K^2 + dK/dx4)\,\hat{\psi}$, in which $dK/dx4$ is a Hermitian matrix with the square $\sum_t (\mathrm{rate}_t\, k_t/b_t)^2$: a Klein–Gordon equation with one additional term that depends on the spin (check `44d-D3`).

3.5 The two coupled 8-component equations

The chiral decompositions of the main project survive, because they are properties of the frame matrices.

4+4. With $\gamma^A_{1644} = \begin{pmatrix} 0 & \bar{\tau}^A \\ \tau^A & 0 \end{pmatrix}$ and $\hat{\psi} = (\hat{\psi}^L, \hat{\psi}^R)$, the equation of the weighted field is the pair of real 8-component equations

$$\bar{\tau}^A\, e_A{}^\mu \partial_\mu\, \hat{\psi}^R = m\,\hat{\psi}^L, \qquad \tau^A\, e_A{}^\mu \partial_\mu\, \hat{\psi}^L = m\,\hat{\psi}^R, \qquad e_A{}^\mu \partial_\mu = \Big(\partial_0, \dots, \partial_4, \frac{\partial_5}{b_5}, \frac{\partial_6}{b_6}, \frac{\partial_7}{b_7}\Big) \tag{3.11}$$

(check `44d-D6`).

7+1. With the complex 16×8 matrices E^L, E^R of the main project and $\hat{\varphi}^{L,R} = (E^{L,R})^\dagger \hat{\psi}$,

$$(i\,\partial_4 + \tau^j\, e_j{}^\mu \partial_\mu)\hat{\varphi}^L = m\,\hat{\varphi}^R, \qquad (i\,\partial_4 - \tau^j\, e_j{}^\mu \partial_\mu)\hat{\varphi}^R = m\,\hat{\varphi}^L, \tag{3.12}$$

where j runs over the seven space-like directions $0, 1, 2, 3, 5, 6, 7$ and $\tau^j = \tau^j_{1671}$ (check `71d-D6`).

3.6 Conserved quantities and charge conjugation

Current. With the adjoint $\bar{\psi} = \psi^\dagger\gamma^4$ on 7+1 and $\tilde{\psi} = \psi^\dagger C^-$ on 4+4, the spin connection satisfies $\Omega_\mu^T\gamma^4 + \gamma^4\Omega_\mu = 0$ and $\Omega_\mu^T C^- + C^-\Omega_\mu = 0$ respectively (checks `71d-C8`, `44d-C8`). Together with Eq. (2.6) this makes the currents $j^\mu = \bar{\psi}\gamma^\mu\psi$ and $j^\mu = \tilde{\psi}\gamma^\mu\psi$ of the main project conserved on the deflating spacetimes, $\nabla_\mu j^\mu = 0$, which is the same as $\partial_\mu(V_d\, j^\mu) = 0$: the charge is the integral of $V_d\, j^4$ over the seven coordinates other than $x4$.

7+1. $V_d\, j^4 = \hat{\psi}^\dagger\hat{\psi}$: the conserved charge is positive, and it is the number of particles minus the number of antiparticles after quantization, as on the flat spacetime. If ψ is a solution then so is the charge-conjugate field $\omega_{1671}\psi^*$ (check `71d-D5`).

4+4. $V_d\, j^4 = \hat{\psi}^\dagger W_{1644}\hat{\psi}$ with the matrix $W_{1644} = i\,C^-\gamma^4$ of the main project, which is Hermitian, has the square one and the trace zero; $W_{1644}K + K^\dagger W_{1644} = 0$ for all wavenumbers (check `44d-D4`). The conserved charge of the 4+4 field is not positive, on the deflating spacetime as on the flat one.

Part 4. Solutions, mode by mode

4.1 Three kinds of modes

1. **Modes that do not depend on the deflating coordinates** ($k_5 = k_6 = k_7 = 0$). By Sec. 3.3 the weighted field obeys the flat equation. The solutions are the plane waves of `dirac matrices`, with energies $\pm E_\perp$, eight of each sign. Nothing is created and nothing is extinguished.

2. **Modes that depend on a deflating coordinate, early** ($k_d \ll E_\perp$). The physical wavenumber is negligible, and the mode behaves as in case 1.
3. **The same modes, late** ($k_d \gg E_\perp$). The physical wavenumber dominates. This always happens: k_d grows without bound, and every mode with $k_t \neq 0$ crosses from case 2 to case 3 at the moment at which $k_d = E_\perp$.

The crossing is where the deflation acts. The rest of this part solves the two equations (3.7) and (3.9) across it. Write the part of h and of K that contains the scale factors with the help of one matrix,

$$\gamma^d = \frac{1}{k_d}\sum_{t=5}^{7}\frac{k_t}{b_t(x4)}\,\gamma^t, \qquad \alpha^d = \gamma^4\gamma^d: \qquad \mathsf{h} = \mathsf{h}_\perp + k_d\,\alpha^d, \qquad K = K_\perp + i\,k_d\,\gamma^4\gamma^d, \tag{4.1}$$

where $\mathsf{h}_\perp = m\beta + \sum_a p_a\alpha^a$ and $K_\perp = -m\gamma^4 + i\sum_a k_a\gamma^4\gamma^a$ are the operators of the modes that do not depend on the deflating coordinates. γ^d is the frame matrix along the direction of the wave vector in the deflating coordinates; $(\gamma^d)^2 = -1$. The solution is exact when *one rate governs*, that is, when only one k_t differs from zero or when the three rates are equal. In those cases γ^d does not depend on $x4$ and $k_d(x4)$ is proportional to $e^{\mathrm{rate}_t\,x4}$. The natural variable is then

$$u_d = \frac{k_d(x4)}{\mathrm{rate}_t}, \qquad \frac{d}{dx4} = \mathrm{rate}_t\,u_d\,\frac{d}{du_d}, \tag{4.1a}$$

the physical wavenumber in units of the rate. The crossing is at $u_d = E_\perp/\mathrm{rate}_t$; this ratio of the energy to the rate is the one parameter on which the results below depend.

4.2 7+1: the deflation creates particles

Exact solution. Let one rate govern. In $\mathsf{h} = \mathsf{h}_\perp + k_d(x4)\,\alpha^d$ the two constant matrices satisfy $\mathsf{h}_\perp^2 = E_\perp^2$, $(\alpha^d)^2 = 1$ and $\mathsf{h}_\perp\alpha^d + \alpha^d\mathsf{h}_\perp = 0$. Let w be one of the eight eigenvectors of $\mathsf{h}_\perp$ with the eigenvalue $+E_\perp$; then $\alpha^d w$ is an eigenvector with the eigenvalue $-E_\perp$. The ansatz $\hat{\psi} = f_+(x4)\,w + i\,f_-(x4)\,\alpha^d w$ turns Eq. (3.7) into two equations for two functions,

$$f_+' = -iE_\perp f_+ + k_d\,f_-, \qquad f_-' = -k_d\,f_+ + iE_\perp f_-.$$

In the variable u_d these are solved by Bessel functions whose orders are $\pm\frac{1}{2} + iE_\perp/\mathrm{rate}_t$. Let Z_+ and Z_- denote two cylinder functions of the same kind (Bessel functions of the first kind, of the second kind, or Hankel functions) of the orders $\frac{1}{2}+iE_\perp/\mathrm{rate}_t$ and $-\frac{1}{2}+iE_\perp/\mathrm{rate}_t$. The recurrence relations of the cylinder functions connect two functions whose orders differ by one; for this pair they read, with a prime for the derivative with respect to u_d,

$$u_d\,Z_+' = u_d\,Z_- - \Big(\frac{1}{2} + \frac{iE_\perp}{\mathrm{rate}_t}\Big)Z_+, \qquad u_d\,Z_-' = -u_d\,Z_+ + \Big(-\frac{1}{2} + \frac{iE_\perp}{\mathrm{rate}_t}\Big)Z_-.$$

With them, substitution shows that

$$\boxed{\hat{\psi} = \sqrt{u_d}\,\Big[Z_+(u_d)\;w + i\,Z_-(u_d)\;\alpha^d w\Big], \qquad \text{orders of } Z_\pm:\;\; \pm\frac{1}{2} + \frac{iE_\perp}{\mathrm{rate}_t}} \tag{4.2}$$

is a solution for each of the eight w; two independent kinds of cylinder function give all sixteen solutions. Check `I71d-M1` verifies Eq. (4.2) with the Bessel functions of an arbitrary-precision library to a relative residual of 10^{-15}.

Number of created particles. At early times ($u_d \to 0$) a Bessel function of the first kind behaves as the power of its argument that its order gives. The solution that has positive energy at early

times, $\propto e^{-iE_\perp x4}\,w$, is therefore Eq. (4.2) with the Bessel functions of the first kind of the *negative* orders: Z_+ is the function of the order $-\frac{1}{2}-iE_\perp/\mathrm{rate}_t$ and Z_- is minus the function of the order $\frac{1}{2}-iE_\perp/\mathrm{rate}_t$, a pair that obeys the same recurrence relations. At late times ($u_d\to\infty$) the solution of positive energy is Eq. (4.2) with the Hankel functions of the second kind, which behave as e^{-iu_d}, and the solution of negative energy is Eq. (4.2) with the Hankel functions of the first kind. The connection formula of the Bessel functions writes the first solution as a sum of the other two, with coefficients of the magnitudes $\frac{1}{2}e^{+\pi E_\perp/\mathrm{rate}_t}$ (for the Hankel solution of the second kind) and $\frac{1}{2}e^{-\pi E_\perp/\mathrm{rate}_t}$ (first kind). At large argument the squared magnitudes of the two Hankel solutions themselves are $(2/\pi)\,e^{-\pi E_\perp/\mathrm{rate}_t}$ (second kind) and $(2/\pi)\,e^{+\pi E_\perp/\mathrm{rate}_t}$ (first kind). The weight of the negative-energy part, divided by the weight of the positive-energy part, is therefore $e^{-2\pi E_\perp/\mathrm{rate}_t}$. A mode that starts empty ends with

$$\boxed{n_{\mathrm{cr}}=\frac{1}{e^{2\pi E_\perp/\mathrm{rate}_t}+1}} \tag{4.3}$$

particles in each of its eight internal states, and with as many antiparticles.

Meaning. Equation (4.3) is the Fermi–Dirac distribution in the energy $E_\perp$ at the temperature

$$T_{\mathrm{defl}}=\frac{\mathrm{rate}_t}{2\pi} \tag{4.3a}$$

(in units with $\hbar=k_B=1$). For equal rates this is the Gibbons–Hawking temperature of de Sitter space. Each mode with $k_t\neq 0$ is populated once, at its crossing, and its particles then gain energy without bound, $\sqrt{E_\perp^2+k_d^2}$, as the classical particle of Sec. 1.7 does. Modes with $k_t=0$ stay empty.

Numerical verification. The program integrates Eq. (3.7) for all sixteen components through the crossing and projects the result on the states of negative energy.

- *One direction* (check `71d-M1`): for $E_\perp/\mathrm{g55}=0.25,0.5,\ldots,3.0$ the created number agrees with Eq. (4.3) to a relative deviation of at most 3.2×10^{-6}, while the numbers themselves range from 0.17 to 6.5×10^{-9}; the eight internal states are created equally.
- *Equal rates, all three directions* (check `71d-M2`): for a mode with $(k_5,k_6,k_7)=(0.5,-0.8,0.3)$ and three different constants the created number is 5.70627486×10^{-3} in all eight states, and Eq. (4.3) gives 5.70627482×10^{-3}. With equal rates the result depends neither on the direction of the wavenumber nor on the constants.
- *Unequal rates* (check `71d-M3`): no exact solution is known. If the second physical wavenumber is still below $e^{-3}E_\perp$ at the moment at which the first reaches $E_\perp$, the mean created number is Eq. (4.3) with the rate of the coordinate that becomes energetic first, to within 1.2 per cent. If two directions become energetic together, the eight internal states are created *unequally* — in the cases computed, by up to a factor 110 between the most and the least populated state — and the result is known only numerically (`data/creation_anisotropic.csv`).
- *Independent integration* (checks `I71d-M2`, `I71d-M3`): a Runge–Kutta method of a general-purpose library reproduces Eq. (4.3) to 2.2×10^{-5}.

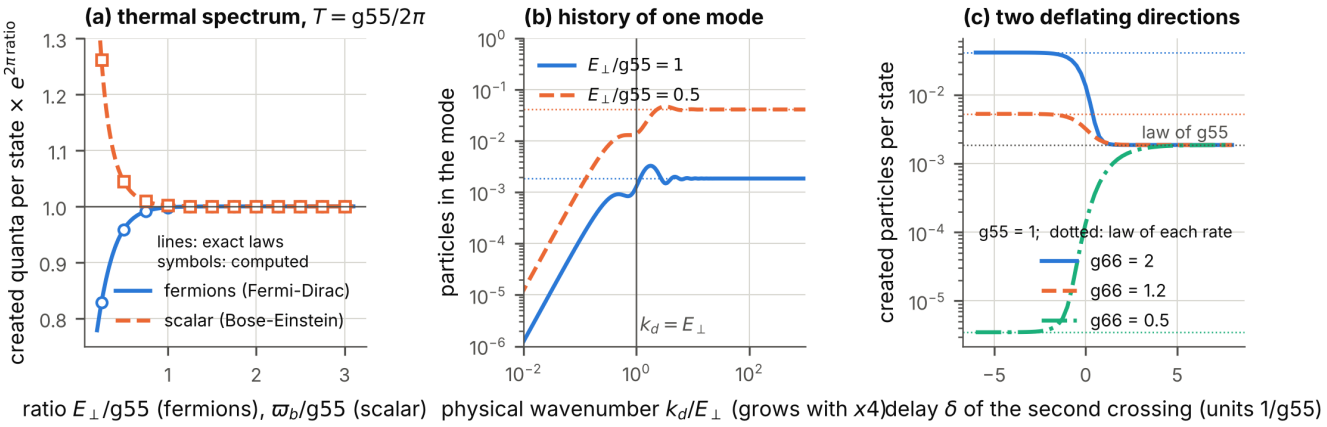


Figure 3: Creation of particles on the deflating 7+1 spacetime. (a) Created quanta per state, multiplied by $e^{2\pi \text{ ratio}}$, against the ratio of the energy to the rate: fermions follow the Fermi–Dirac law (4.3), the scalar modes of the supplement RS1 the Bose–Einstein law, both at the temperature g55$/2\pi$. Lines: the exact laws; symbols: computed. (b) History of two modes, with $E_\perp/\text{g55} = 1$ and 0.5: the number of particles in the mode against the physical wavenumber $k_d/E_\perp$, which grows with the time; after the crossing at $k_d = E_\perp$ the number settles at the value of Eq. (4.3) (dotted). (c) Two deflating directions, g55 $= 1$ and g66 $= 2$, 1.2 or 0.5: mean number of created particles per state against the delay between the two crossings; the dotted lines are Eq. (4.3) with each of the rates. Data: `data/creation_single.csv`, `data/boson_creation.csv`, `data/creation_history.csv`, `data/creation_anisotropic.csv`; programs: `deflation modes`, `deflation channels`.

4.3 4+4: the deflation extinguishes the dependence on the deflating times

Exact solution. Let one rate govern. In $K = K_\perp + i\,k_d(x4)\,\gamma^4\gamma^d$ the first matrix is anti-Hermitian with $K_\perp^2 = -E_\perp^2$; the matrix $i\gamma^4\gamma^d$ is Hermitian with the square one; and the two anticommute. Let w be one of the eight eigenvectors of $K_\perp$ with the eigenvalue $+iE_\perp$; then $i\gamma^4\gamma^d w$ has the eigenvalue $-iE_\perp$ and is orthogonal to w. The ansatz $\hat{\psi} = f_+\,w + f_-\,i\gamma^4\gamma^d w$ gives

$$f_+' = iE_\perp f_+ + k_d\,f_-, \qquad f_-' = k_d\,f_+ - iE_\perp f_-.$$

These are solved by the modified Bessel functions that decay at large argument, of the orders $\frac{1}{2} \mp iE_\perp/\text{rate}_t$. Such a function is written with its order as a subscript and with an argument; the letter K without subscript and argument is the matrix of Eq. (3.9). The recurrence relation for this pair of orders is

$$u_d\,K'_{1/2-iE_\perp/\text{rate}_t} = -u_d\,K_{1/2+iE_\perp/\text{rate}_t} - \left(\frac{1}{2} - \frac{iE_\perp}{\text{rate}_t}\right) K_{1/2-iE_\perp/\text{rate}_t},$$

together with the relation obtained from it by reversing the sign of i. With them, substitution shows that

$$\boxed{\hat{\psi} = \sqrt{u_d}\left[K_{1/2-iE_\perp/\text{rate}_t}(u_d)\;w - K_{1/2+iE_\perp/\text{rate}_t}(u_d)\;i\gamma^4\gamma^d w\right]} \tag{4.4}$$

is a solution for each of the eight w (check `I44d-M1`, relative residual 10^{-15}). The other eight solutions are built in the same way from the modified Bessel functions that *grow* as e^{+u_d}.

The bounded solutions. Since u_d itself grows exponentially with $x4$, a solution of the second kind grows as the exponential of an exponential. Such a solution is not admissible: it is the analogue of the growing exponential under a potential barrier. Admissible are the eight solutions (4.4), which stay bounded. Their properties are exact.

- *Magnitude.* Because w and $i\gamma^4\gamma^d w$ are orthogonal, and because the two functions in Eq. (4.4) are complex conjugates of each other for real argument,

$$|\widehat{\psi}|^2 = 2\,u_d\,\left|K_{1/2+iE_\perp/\mathrm{rate}_t}(u_d)\right|^2\,|w|^2. \tag{4.5}$$

  The numerical integration reproduces Eq. (4.5) over the range $0.02 < u_d < 12$ to a relative deviation of 3.6×10^{-12} (check `44d-M4`).
- *Early times* ($u_d \to 0$): $|\widehat{\psi}|^2 \to \pi\,|w|^2/\cosh(\pi E_\perp/\mathrm{rate}_t)$, a constant. The solution is then a combination of $e^{+iE_\perp x4}\,w$ and $e^{-iE_\perp x4}\,i\gamma^4\gamma^d w$ with coefficients of *equal* magnitude: the two signs of the frequency enter with equal weight. A bounded solution is a **standing wave** in the ordinary time, not a travelling wave (check `44d-M1`: the parts of positive and of negative energy have equal weight to 2×10^{-14}). Its conserved charge $\widehat{\psi}^\dagger W \widehat{\psi}$ vanishes (check `44d-M2`).
- *Late times* ($u_d \to \infty$): $|\widehat{\psi}|^2 \to \pi\,|w|^2\,e^{-2u_d}$. The mode dies out as the exponential of an exponential of the time.

The law of the extinction. In the semiclassical (WKB) approximation the bounded solution falls beyond the turning value $x4_{\mathrm{turn}}$, at which $k_d = E_\perp$, as $e^{-S_{\mathrm{WKB}}}$ with

$$S_{\mathrm{WKB}} = \int_{x4_{\mathrm{turn}}}^{x4} \sqrt{k_d^2 - E_\perp^2}\;dx4' = \frac{1}{\mathrm{rate}_t}\Big[\sqrt{k_d^2 - E_\perp^2} - E_\perp \arccos\frac{E_\perp}{k_d}\Big] \;\longrightarrow\; \frac{k_d - \frac{\pi}{2}E_\perp}{\mathrm{rate}_t} \quad (k_d \gg E_\perp).$$

The exact solution (4.5) fixes the constant that the approximation leaves open:

$$\ln\frac{|\widehat{\psi}|_{\mathrm{early}}}{|\widehat{\psi}|_{\mathrm{late}}} - S_{\mathrm{WKB}} \;\longrightarrow\; \tfrac{1}{2}\ln 2 - \tfrac{1}{2}\ln\left(1 + e^{-2\pi E_\perp/\mathrm{rate}_t}\right). \tag{4.6}$$

The numerical integration confirms Eq. (4.6) to 3.6×10^{-4} in the four cases in which one rate governs; with three unequal rates the constant depends on the internal state, with a spread of 0.40 at $S_{\mathrm{WKB}} = 30$ (check `44d-M3`). The independent script confirms Eq. (4.6) with the Bessel functions of another library (check `I44d-M2`).

Meaning. On the flat 4+4 spacetime a field that depends on the additional times evolves non-unitarily in the ordinary time, which is the principal difficulty of four time-like dimensions (Sec. 1.8.4 of `dirac matrices`). On the deflating 4+4 spacetime every admissible mode that depends on a deflating time is a standing wave that is extinguished after its crossing, with S_{WKB} growing as $e^{\mathrm{rate}_t\,x4}$. What remains at late times is the sector of fields that do not depend on $x5$, $x6$, $x7$ — and in that sector the evolution is unitary and $\widehat{\psi}^\dagger\widehat{\psi}$ is conserved (check `44d-D5`). The deflation of the three additional times thus removes, dynamically, the sector that causes the difficulty. The classical counterpart is the turning value of Sec. 1.7. Part 6 states the limits of this conclusion.

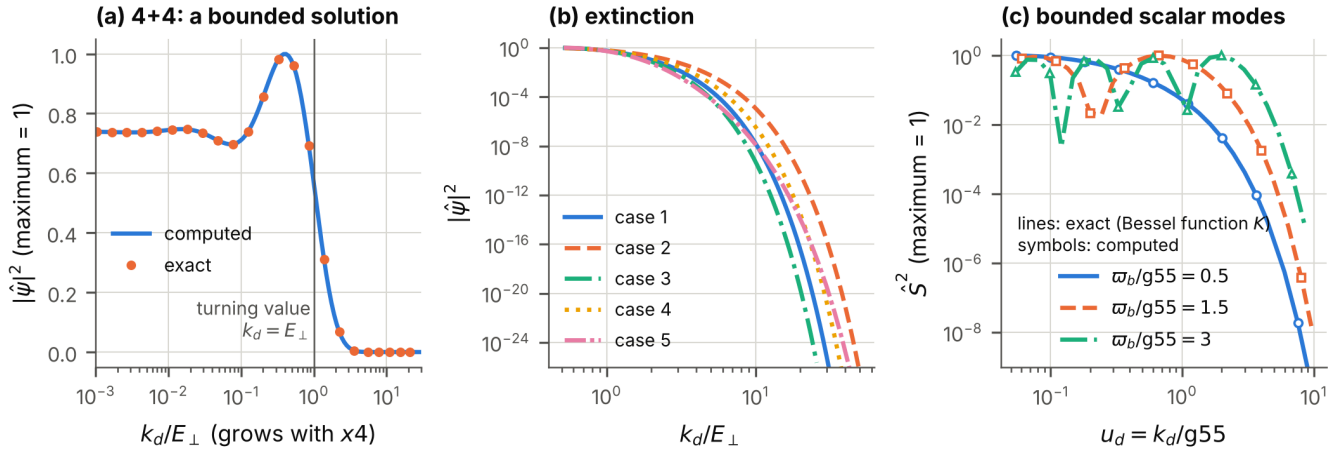


Figure 4: Extinction on the deflating 4+4 spacetime. (a) A bounded solution: $|\hat\psi|^2$ against the physical wavenumber $k_d/E_\perp$, which grows with the time; computed (line) and the exact form (4.5) (symbols). It is constant long before the crossing at $k_d = E_\perp$ and falls after it. (b) The five cases of check `44d-M3` on a logarithmic scale. (c) The bounded modes of a scalar field on the same spacetime (supplement RS1), which are extinguished in the same way; lines: exact, symbols: computed. Data: `data/extinction_profile.csv`, `data/extinction_checks.csv`, `data/boson_bounded.csv`; programs: `deflation modes`, `deflation channels`.

4.4 The numerical method and its tests

Equations (3.7) and (3.9) are linear equations with coefficients that grow exponentially, and a general-purpose integrator with a fixed step fails on them. The program uses a commutator-free Magnus method of order four: over each step the solution is advanced by the product of two matrix exponentials of linear combinations of the coefficient matrix at two Gauss points. Because h^2 and K^2 are multiples of the unit matrix at any one time, each exponential is evaluated in closed form (a cosine and a sine, or their hyperbolic counterparts), and for h every step is exactly unitary. The length of each step is chosen so that neither the phase accumulated in the step nor the relative change of the coefficients exceeds a fixed bound; the steps therefore shorten as the coefficients grow.

Reading off the number of particles. At a finite time the state of positive energy of the instantaneous Hamiltonian is not yet the state of a free particle, because the Hamiltonian still changes. The program projects instead on the eigenstates of the first "superadiabatic" Hamiltonian, which is h corrected by a term of the first order in the rate at which the direction of h changes (the formula is in the documentation of the function `negative_projector` of `modes.rs`). This removes the leading spurious oscillation of the particle number. The error that remains decreases as the final value of $k_d/E_\perp$ is increased, and is contained in the deviations quoted above.

Tests. The unit tests of the modules `modes` and `scalar` (run by `cargo test`) verify the order of convergence of the method, the unitarity of every step, the agreement with Eq. (4.3) and with Eq. (4.5), and the modified Bessel function of complex order, which the program computes from its integral representation. The independent script uses different tools throughout: an explicit Runge–Kutta method of order eight and the Bessel functions of an arbitrary-precision library.

Part 5. Which symmetries of the flat spacetimes survive

Rotations and boosts. The flat spacetimes have 28 rotations and boosts, one for each pair of coordinates. The program tests each of the 28 on the deflating spacetime (checks `44d-S1`, `71d-S1`):

- the 6 rotations of $x0, x1, x2, x3$ survive for any rates;

- the rotation of two deflating coordinates survives if and only if their rates are equal (in the form given in Sec. 1.5, which contains the two constants);
- nothing that involves $x4$ survives — on 7+1 these are the seven boosts, among them the three boosts of ordinary special relativity along $x1, x2, x3$;
- nothing that mixes $x0$, or any flat coordinate, with a deflating coordinate survives.

That the boosts along $x1, x2, x3$ are no longer symmetries deserves a remark. The deflation defines a preferred state of rest, as the expansion of the universe does: the frame in which the scale factors depend on the time alone and not on the position. For fields that do not depend on $x5, x6, x7$ the flat equations hold for the weighted field (Sec. 3.3), and their solutions are related by the ordinary boosts as before. What singles out the preferred frame for such fields is only the dependence of the couplings on $x4$ (supplement RS1).

The internal symmetry of fields that do not depend on the extra coordinates. The main project found that the eight internal states of the fermion, for fields that depend on $x1, x2, x3, x4$ only, carry the symmetry $su(2)_A \oplus su(2)_B \oplus u(1)_F \oplus u(1)_E$. Because the weighted field of this sector obeys the flat equations, the symmetry is unchanged by the deflation (check `def-S3`). Six of its generators are the products $\gamma^A\gamma^B$ of the six pairs taken from the four axes $x0, x5, x6, x7$; on the flat spacetime they are rotations of these axes. On the deflating spacetime 0, 1 or 3 of the six are still isometries, for three different, two equal or three equal rates (checks `44d-S2`, `71d-S2`). The others remain exact symmetries of the sector *without* being symmetries of the spacetime. The supplement RS2 examines what this means for the hypothesis about the Higgs field.

Discrete symmetries. The scale factors are real and depend on $x4$ only. Reflections of any coordinate other than $x4$ and charge conjugation (check `71d-D5`) survive. The reversal of the ordinary time does not: it turns deflation into inflation.

Part 6. What has been established, and what has not

Established (each item by exact arithmetic at rational parameters and, where marked with I, symbolically for arbitrary parameters).

1. The geometry of the two deflating spacetimes: Christoffel symbols, Riemann and Ricci tensors, curvature scalar and Einstein tensor in closed form (I); the Killing vectors, 14, 15 or 20, and the completeness of the lists; homogeneity; constant curvature of the four coordinates $x4, \dots, x7$ for equal rates.
2. The deflating spacetimes are not vacuum solutions of Einstein's equations, with or without a cosmological constant. With three equal rates they are exact vacuum solutions of Einstein–Gauss–Bonnet gravity, for $\Lambda_8 = 3\,\mathrm{g55}^2$ and $\alpha_{GB} = -1/(4\,\mathrm{g55}^2)$ (I); with unequal rates they are not.
3. Geodesics: unbounded blueshift and geodesic incompleteness on 7+1; a turning value of the ordinary time on 4+4.
4. The Dirac matrices in coordinate components, the spin connection, the covariant constancy of the matrices, and the curvature of the spinor derivative (I).
5. The Dirac equations (3.1) and (3.3); the weighted field; the statement that fields independent of $x5, x6, x7$ obey the flat equations; the operators h and K and their squares (I); the 8-component forms; the conserved currents.
6. The exact solutions (4.2) and (4.4) when one rate governs, the law of creation (4.3) on 7+1, and the standing waves and the law of extinction (4.5), (4.6) on 4+4; all confirmed numerically by two independent methods.
7. Which rotations and boosts survive, and that the internal symmetry of the sector of fields independent of the extra coordinates does not change.

Not established.

1. *The source of the deflation.* For equal rates Sec. 1.4a gives an answer within Einstein–Gauss–Bonnet gravity; its stability has not been examined. For unequal rates no field equation has

been found that the metric solves, and the stress tensor that Eq. (1.10) would require has not been identified with any matter. All results on fields are results on a prescribed background.

2. *The effect of the fields on the metric.* The particles created on 7+1 gain energy without bound. Their stress tensor must eventually change the metric; this back-reaction has not been computed, and the results for 7+1 hold only for as long as it can be neglected.
3. *The future boundary of 7+1.* The coordinates do not cover the whole future of the particles that move along deflating coordinates (Sec. 1.7). What lies beyond is outside the hypotheses.
4. *The exclusion of the growing solutions on 4+4.* The eight growing solutions are discarded because they are unbounded. That is a condition of admissibility, imposed as in the theory of a barrier; it is not derived from a principle, and the 4+4 theory still has no positive conserved charge for modes that depend on the deflating times. The statement of Sec. 4.3 is that the admissible modes die out, not that the difficulty of the four time-like dimensions has been solved.
5. *Unequal rates with simultaneous crossings.* No closed form is known; the results are numerical.
6. *Quantization on 4+4.* As in the main project, a consistent quantization of the 4+4 field exists only for the sector that does not depend on the additional times.

Appendix A. Complete output of the programs

In all files “exact” means that the statement is an identity verified in exact rational arithmetic, and “numerical” that it involves real numbers and is verified to the accuracy stated in the line.

A.1 Geometry, matrices, spin connection, Dirac operators

The file `data/deflating_checks.txt`, written by `deflation matrices`:

```
PROJECT spacetimes, Revision_Deflation: checks of the geometry, of the Dirac matrices, of the spin
 ↪  connection and of the Dirac operators of the deflating spacetimes (44d-..: 4+4 with deflating
 ↪  x5, x6, x7; 71d-..: 7+1 with deflating x5, x6, x7)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation matrices
47 of 47 checks passed

[PASS] 44d-G1  (exact) metric of the deflating 4+4 spacetime: g_1644d = diag(eta_00, .., eta_44,
 ↪  eta_55 b_5^2, eta_66 b_6^2, eta_77 b_7^2) with eta = diag[1, 1, 1, 1, -1, -1, -1, -1] and the
 ↪  scale factors b_5 = const5 exp(-g55 x4), b_6 = const6 exp(-g66 x4), b_7 = const7 exp(-g77 x4),
 ↪  that is g_55 = -const5^2 exp(-2 g55 x4) and likewise g_66, g_77 (the author's formulas; g55,
 ↪  g66, g77 > 0 are the three rates, and in statements with a general index t = 5, 6, 7 the rate
 ↪  of xt is written rate_t, gsum = g55 + g66 + g77). Verified: g_tt = -b_t^2 and d g_tt/dx4 = -2
 ↪  rate_t g_tt; the frame e^A = (dx0, .., dx4, b_5 dx5, b_6 dx6, b_7 dx7) is orthonormal. All
 ↪  exact geometry checks use five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a point
 ↪  where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5, b_6,
 ↪  b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6, b_7) =
 ↪  (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2, 3/2, 1/4);
 ↪  (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)
[PASS] 44d-G2  (exact) the Christoffel symbols computed from their definition are Gamma^t_(4t) =
 ↪  Gamma^t_(t4) = -rate_t and Gamma^4_(tt) = eta_44 eta_tt rate_t b_t^2 (t = 5, 6, 7); all others
 ↪  vanish
[PASS] 44d-G3  (exact) Cartan's structure equations hold with the spin-connection coefficients of
 ↪  the tetrad postulate: the torsion d e^A + omega^A_B ^ e^B vanishes, and d omega^A_B +
 ↪  omega^A_C ^ omega^C_B equals the Riemann tensor computed from the Christoffel symbols (two
 ↪  independent routes to the curvature)
[PASS] 44d-G4  (exact) Riemann tensor in the orthonormal frame: R^t_(4t4) = -rate_t^2, R^4_(t4t) =
 ↪  -eta_44 eta_tt rate_t^2, R^t_(sts) = -eta_44 eta_ss rate_t rate_s (t != s in {5, 6, 7}), and
 ↪  the components that follow from antisymmetry; all other components vanish. No component
 ↪  involves x0, x1, x2 or x3: the space of these four coordinates is flat
```

```
[PASS] 44d-G5  (exact) with gsq = g55^2 + g66^2 + g77^2 and gsum = g55 + g66 + g77: Ricci tensor
 ↪  R^4_4 = -eta_44 gsq, R^t_t = -eta_44 rate_t gsum, R^a_a = 0 (a = 0..3); curvature scalar R =
 ↪  -eta_44 (gsq + gsum^2); Einstein tensor G^4_4 = eta_44 (g55 g66 + g55 g77 + g66 g77), G^t_t =
 ↪  eta_44 [(gsq + gsum^2)/2 - rate_t gsum], G^a_a = eta_44 (gsq + gsum^2)/2; its covariant
 ↪  divergence vanishes. The Einstein tensor does not vanish: the deflating spacetime is not a
 ↪  vacuum solution of Einstein's equations
[PASS] 44d-G6  (exact) Killing vectors (the Lie derivative of the metric along each vanishes
 ↪  identically): 7 translations (x0, x1, x2, x3, x5, x6, x7), 6 rotations of (x0, x1, x2, x3),
 ↪  and the scaling d/dx4 + sum_t rate_t xt d/dxt, 14 in all; one more for every pair of equal
 ↪  rates (a rotation of the two coordinates); and 20 in all if the three rates are equal. Numbers
 ↪  found for the five parameter sets: [20, 14, 14, 15, 20]
[PASS] 44d-G7  (exact) if the three rates are equal, g55 = g66 = g77, the part (x4, x5, x6, x7) of
 ↪  the metric has constant curvature, R_(rho sigma mu nu) = +g55^2 (g_(rho mu) g_(sigma nu) -
 ↪  g_(rho nu) g_(sigma mu)): the four time-like coordinates carry minus the metric of hyperbolic
 ↪  space H^4 of radius 1/g55, written in horospherical coordinates
[PASS] 44d-G10  (exact) Lovelock gravity. In the convention in which the space-like components of
 ↪  the metric are positive (the metric g_1644d itself), with the action R - 2 Lambda_8 + alpha_GB
 ↪  (R^2 - 4 R_(mu nu) R^(mu nu) + R_(mu nu rho sigma) R^(mu nu rho sigma)) and the field
 ↪  equations G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB Lanczos^mu_nu = 0: the curvature acts
 ↪  diagonally on pairs of frame directions; the Lanczos tensor of the Gauss-Bonnet term is
 ↪  Lanczos^a_a = -4 g55 g66 g77 gsum (a = 0..3), Lanczos^4_4 = Lanczos^t_t = 0; the Lovelock
 ↪  tensor of third order vanishes identically. For three equal rates (2 of the five parameter
 ↪  sets) the deflating spacetime solves the vacuum field equations with Lambda_8 = 3 g55^2 and
 ↪  alpha_GB = -1/(4 g55^2), that is 4 alpha_GB Lambda_8 = -3: the deflation needs no matter as
 ↪  its source in Einstein-Gauss-Bonnet gravity. For rates that are not all equal no choice of
 ↪  Lambda_8 and alpha_GB satisfies the components 4, 5, 6, 7
[PASS] 44d-G9  (numerical) geodesic with m = 1, momentum P_1 = 0.5 along x1 and P_5 = 0.3 along
 ↪  the deflating time x5 (g55 = 1, const5 = 0.8), started at x4 = -2 with increasing x4: P_1, P_5
 ↪  and the norm are conserved to 1.7e-14; the path reaches the largest value x4 = 1.09240103
 ↪  (predicted turning value ln(E_perp const5/|P_5|)/g55 with E_perp = sqrt(m^2 + P_1^2) =
 ↪  1.09240103) after the affine parameter 3.38543615 (closed form 3.38543615) and then returns to
 ↪  smaller x4: a particle with momentum along a deflating time cannot pass the turning value
[PASS] 44d-C1  (exact) the coordinate matrices gamma_1644d^mu = e_A^mu gamma1644[A] (gamma^t =
 ↪  gamma1644[t]/b_t for t = 5, 6, 7; the other five unchanged) are real and satisfy
 ↪  gamma^mu.gamma^nu + gamma^nu.gamma^mu = 2 g_1644d^(mu nu) 1 for all 64 pairs, with g^(tt) =
 ↪  eta^(tt)/b_t^2. Verified for five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a
 ↪  point where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5,
 ↪  b_6, b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6,
 ↪  b_7) = (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2,
 ↪  3/2, 1/4); (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)
[PASS] 44d-C2  (exact) the spin-connection coefficients computed from the tetrad postulate
 ↪  omega_mu^A_B = e^A_nu (d_mu e_B^nu + Gamma^nu_(mu lam) e_B^lam), with the Christoffel symbols
 ↪  obtained from the metric, are antisymmetric, omega_(mu A B) = -omega_(mu B A), and the only
 ↪  non-zero ones are omega_(t,t4) = -omega_(t,4t) = -eta_tt rate_t b_t (t = 5, 6, 7); same five
 ↪  parameter sets
[PASS] 44d-C3  (exact) Omega_mu := (1/4) omega_(mu A B) gamma1644[A].gamma1644[B] equals -(1/2)
 ↪  eta_tt rate_t b_t gamma1644[t].gamma1644[4] for mu = t in {5, 6, 7} and vanishes for mu =
 ↪  0..4: the spin connection is an element of the Lie algebra spanned by the scaled commutators
 ↪  Sigma^(t4)
[PASS] 44d-C4  (exact) d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for
 ↪  all 64 pairs (mu, nu), with d_4 gamma^t = +rate_t gamma^t: the coordinate matrices are
 ↪  covariantly constant
[PASS] 44d-C5  (exact) sum_mu gamma^mu.Omega_mu = -(1/2) (g55 + g66 + g77) gamma1644[4]: in the
 ↪  Dirac operator the spin connection amounts to the replacement d_4 -> d_4 - gsum/2, which the
 ↪  weighted field psihat = sqrt(b_5 b_6 b_7) psi removes
[PASS] 44d-C6  (exact) d_mu Omega_nu - d_nu Omega_mu + [Omega_mu, Omega_nu] = (1/4) R_(A B mu nu)
 ↪  gamma1644[A].gamma1644[B] for all 64 pairs (mu, nu), with the Riemann tensor that metric.rs
 ↪  computes from the metric (d_4 Omega_t = -rate_t Omega_t): the curvature of the spinor
 ↪  derivative is the Riemann tensor
```

```
[PASS] 44d-C7  (exact) (1/2) sum_(mu nu) gamma^mu.gamma^nu.(d_mu Omega_nu - d_nu Omega_mu +
↪  [Omega_mu, Omega_nu]) = -(R/4) 1 with the curvature scalar R = -eta_44 (g55^2 + g66^2 + g77^2
↪  + gsum^2): the square of the Dirac operator differs from the wave operator of the spinor
↪  derivative by -R/4
[PASS] 44d-C8  (exact) transpose(Omega_mu).Cm1644 + Cm1644.Omega_mu = 0 (Cm1644 =
↪  gamma1644[0].gamma1644[1].gamma1644[2].gamma1644[3]): the adjoint field psi^dagger Cm1644 is
↪  differentiated with -Omega_mu on its right, as the conserved current requires
[PASS] 44d-D1  (exact) in momentum space (d_mu -> i k_mu) the Dirac operator of the weighted field
↪  at fixed x4, D = i gamma^mu k_mu with the coordinate matrices, satisfies (D + m)(D - m) =
↪  -(g^(mu nu) k_mu k_nu + m^2) 1 with g^(mu nu) k_mu k_nu = k_0^2 + k_1^2 + k_2^2 + k_3^2 -
↪  k_4^2 - sum_t k_t^2/b_t^2: the instantaneous mass shell is k_4^2 + sum_t k_t^2/b_t(x4)^2 =
↪  |k|^2 + m^2. Verified for five rational parameter sets and four rational momenta each
[PASS] 44d-D2  (exact) the generator of the evolution of the weighted field in x4, K(x4) = -m
↪  gamma1644[4] + i sum_a k_a gamma1644[4].gamma1644[a] + i sum_t (k_t/b_t(x4))
↪  gamma1644[4].gamma1644[t], satisfies K^2 = (sum_t k_t^2/b_t^2 - m^2 - |k|^2) 1 and K +
↪  K^dagger = 2 i sum_t (k_t/b_t) gamma1644[4].gamma1644[t]
[PASS] 44d-D3  (exact) dK/dx4 = i sum_t rate_t (k_t/b_t) gamma1644[4].gamma1644[t] is a Hermitian
↪  matrix whose square is sum_t (rate_t k_t/b_t)^2 times the unit matrix; the weighted field
↪  obeys the second-order equation d_4^2 psihat = (K^2 + dK/dx4) psihat, a Klein-Gordon equation
↪  with a term that depends on the spin
[PASS] 44d-D4  (exact) W = i Cm1644.gamma1644[4] is Hermitian with W^2 = 1 and trace 0, and W.K +
↪  K^dagger.W = 0 for every momentum and all k_5, k_6, k_7: the covariant charge psihat^dagger W
↪  psihat of a mode is conserved on the deflating spacetime, and it is not positive
[PASS] 44d-D5  (exact) K is anti-Hermitian (the evolution in x4 is unitary and psihat^dagger
↪  psihat is conserved) if and only if k_5 = k_6 = k_7 = 0, that is, if the field does not depend
↪  on the deflating times
[PASS] 44d-D6  (exact) with gamma1644[A] = [[0, taubar1644[A]], [tau1644[A], 0]] the operator
↪  gamma^mu d_mu of the weighted field has the blocks [[0, sum_A taubar1644[A] e_A^mu d_mu],
↪  [sum_A tau1644[A] e_A^mu d_mu, 0]], e_A^mu d_mu = (d_0, .., d_4, d_5/b_5, d_6/b_6, d_7/b_7):
↪  the equation is the pair taubar^A e_A^mu d_mu psihat^R = m psihat^L, tau^A e_A^mu d_mu
↪  psihat^L = m psihat^R of real 8-component equations
[PASS] 44d-D7  (exact) worked example: m = 3, k = 0, (k_5, k_6, k_7) = (1, 0, 0) at the value of
↪  x4 where b_5 = 1/5 (so k_5/b_5 = 5 > m): K^2 = +16 1, and (4 + K)/8 and (4 - K)/8 are
↪  complementary projectors of trace 8: at this x4 eight independent solutions grow as exp(+4 x4)
↪  and eight decay as exp(-4 x4) to first order. A mode that depends on a deflating time is
↪  oscillatory only while k_t/b_t(x4) < sqrt(m^2 + |k|^2)
[PASS] 71d-G1  (exact) metric of the deflating 7+1 spacetime: g_1671d = diag(eta_00, .., eta_44,
↪  eta_55 b_5^2, eta_66 b_6^2, eta_77 b_7^2) with eta = diag[-1, -1, -1, -1, 1, -1, -1, -1] and
↪  the scale factors b_5 = const5 exp(-g55 x4), b_6 = const6 exp(-g66 x4), b_7 = const7 exp(-g77
↪  x4), that is g_55 = -const5^2 exp(-2 g55 x4) and likewise g_66, g_77 (the author's formulas;
↪  g55, g66, g77 > 0 are the three rates, and in statements with a general index t = 5, 6, 7 the
↪  rate of xt is written rate_t, gsum = g55 + g66 + g77). Verified: g_tt = -b_t^2 and d g_tt/dx4
↪  = -2 rate_t g_tt; the frame e^A = (dx0, .., dx4, b_5 dx5, b_6 dx6, b_7 dx7) is orthonormal.
↪  All exact geometry checks use five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a
↪  point where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5,
↪  b_6, b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6,
↪  b_7) = (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2,
↪  3/2, 1/4); (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)
[PASS] 71d-G2  (exact) the Christoffel symbols computed from their definition are Gamma^t_(4t) =
↪  Gamma^t_(t4) = -rate_t and Gamma^4_(tt) = eta_44 eta_tt rate_t b_t^2 (t = 5, 6, 7); all others
↪  vanish
[PASS] 71d-G3  (exact) Cartan's structure equations hold with the spin-connection coefficients of
↪  the tetrad postulate: the torsion d e^A + omega^A_B ^ e^B vanishes, and d omega^A_B +
↪  omega^A_C ^ omega^C_B equals the Riemann tensor computed from the Christoffel symbols (two
↪  independent routes to the curvature)
[PASS] 71d-G4  (exact) Riemann tensor in the orthonormal frame: R^t_(4t4) = -rate_t^2, R^4_(t4t) =
↪  -eta_44 eta_tt rate_t^2, R^t_(sts) = -eta_44 eta_ss rate_t rate_s (t != s in {5, 6, 7}), and
↪  the components that follow from antisymmetry; all other components vanish. No component
↪  involves x0, x1, x2 or x3: the space of these four coordinates is flat
```

[PASS] 71d-G5 (exact) with gsq = g55^2 + g66^2 + g77^2 and gsum = g55 + g66 + g77: Ricci tensor R^4_4 = -eta_44 gsq, R^t_t = -eta_44 rate_t gsum, R^a_a = 0 (a = 0..3); curvature scalar R = -eta_44 (gsq + gsum^2); Einstein tensor G^4_4 = eta_44 (g55 g66 + g55 g77 + g66 g77), G^t_t = eta_44 [(gsq + gsum^2)/2 - rate_t gsum], G^a_a = eta_44 (gsq + gsum^2)/2; its covariant divergence vanishes. The Einstein tensor does not vanish: the deflating spacetime is not a vacuum solution of Einstein's equations
[PASS] 71d-G6 (exact) Killing vectors (the Lie derivative of the metric along each vanishes identically): 7 translations (x0, x1, x2, x3, x5, x6, x7), 6 rotations of (x0, x1, x2, x3), and the scaling d/dx4 + sum_t rate_t xt d/dxt, 14 in all; one more for every pair of equal rates (a rotation of the two coordinates); and 20 in all if the three rates are equal. Numbers found for the five parameter sets: [20, 14, 14, 15, 20]
[PASS] 71d-G7 (exact) if the three rates are equal, g55 = g66 = g77, the part (x4, x5, x6, x7) of the metric has constant curvature, R_(rho sigma mu nu) = -g55^2 (g_(rho mu) g_(sigma nu) - g_(rho nu) g_(sigma mu)): it is de Sitter space of radius 1/g55 in the contracting half of its flat slicing
[PASS] 71d-G10 (exact) Lovelock gravity. In the convention in which the space-like components of the metric are positive (minus the metric g_1671d), with the action R - 2 Lambda_8 + alpha_GB (R^2 - 4 R_(mu nu) R^(mu nu) + R_(mu nu rho sigma) R^(mu nu rho sigma)) and the field equations G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB Lanczos^mu_nu = 0: the curvature acts diagonally on pairs of frame directions; the Lanczos tensor of the Gauss-Bonnet term is Lanczos^a_a = -4 g55 g66 g77 gsum (a = 0..3), Lanczos^4_4 = Lanczos^t_t = 0; the Lovelock tensor of third order vanishes identically. For three equal rates (2 of the five parameter sets) the deflating spacetime solves the vacuum field equations with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2), that is 4 alpha_GB Lambda_8 = -3: the deflation needs no matter as its source in Einstein-Gauss-Bonnet gravity. For rates that are not all equal no choice of Lambda_8 and alpha_GB satisfies the components 4, 5, 6, 7
[PASS] 71d-G8 (numerical) geodesic of a particle of mass m = 1 with momentum P_1 = 0.5 along x1 and P_5 = 0.3 along the deflating coordinate x5 (g55 = 1, const5 = 0.8), integrated from x4 = -2 to x4 = 4 by the Runge-Kutta method: P_1, P_5 and the norm are conserved to 2.0e-10; the affine parameter (the proper time, since m = 1) that elapses is 3.3375401839, the closed form gives 3.3375401839; the proper time to x4 = infinity is finite, 3.386358: a particle that moves along a deflating coordinate leaves the coordinate patch after a finite proper time, with unbounded energy
[PASS] 71d-C1 (exact) the coordinate matrices gamma_1671d^mu = e_A^mu gamma1671[A] (gamma^t = gamma1671[t]/b_t for t = 5, 6, 7; the other five unchanged) are real and satisfy gamma^mu.gamma^nu + gamma^nu.gamma^mu = 2 g_1671d^(mu nu) 1 for all 64 pairs, with g^(tt) = eta^(tt)/b_t^2. Verified for five rational parameter sets: (g55, g66, g77) = (1, 1, 1) at a point where (b_5, b_6, b_7) = (1, 1, 1); (g55, g66, g77) = (1, 2, 3) at a point where (b_5, b_6, b_7) = (1/2, 1/3, 1/5); (g55, g66, g77) = (1/2, 3/4, 5/3) at a point where (b_5, b_6, b_7) = (2, 1/3, 5/7); (g55, g66, g77) = (2, 2, 5) at a point where (b_5, b_6, b_7) = (3/2, 3/2, 1/4); (g55, g66, g77) = (7/3, 7/3, 7/3) at a point where (b_5, b_6, b_7) = (1/2, 3, 5/4)
[PASS] 71d-C2 (exact) the spin-connection coefficients computed from the tetrad postulate omega_mu^A_B = e^A_nu (d_mu e_B^nu + Gamma^nu_(mu lam) e_B^lam), with the Christoffel symbols obtained from the metric, are antisymmetric, omega_(mu A B) = -omega_(mu B A), and the only non-zero ones are omega_(t,t4) = -omega_(t,4t) = -eta_tt rate_t b_t (t = 5, 6, 7); same five parameter sets
[PASS] 71d-C3 (exact) Omega_mu := (1/4) omega_(mu A B) gamma1671[A].gamma1671[B] equals -(1/2) eta_tt rate_t b_t gamma1671[t].gamma1671[4] for mu = t in {5, 6, 7} and vanishes for mu = 0..4: the spin connection is an element of the Lie algebra spanned by the scaled commutators Sigma^(t4)
[PASS] 71d-C4 (exact) d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for all 64 pairs (mu, nu), with d_4 gamma^t = +rate_t gamma^t: the coordinate matrices are covariantly constant
[PASS] 71d-C5 (exact) sum_mu gamma^mu.Omega_mu = -(1/2) (g55 + g66 + g77) gamma1671[4]: in the Dirac operator the spin connection amounts to the replacement d_4 -> d_4 - gsum/2, which the weighted field psihat = sqrt(b_5 b_6 b_7) psi removes
[PASS] 71d-C6 (exact) d_mu Omega_nu - d_nu Omega_mu + [Omega_mu, Omega_nu] = (1/4) R_(A B mu nu) gamma1671[A].gamma1671[B] for all 64 pairs (mu, nu), with the Riemann tensor that metric.rs computes from the metric (d_4 Omega_t = -rate_t Omega_t): the curvature of the spinor derivative is the Riemann tensor

```
[PASS] 71d-C7  (exact) (1/2) sum_(mu nu) gamma^mu.gamma^nu.(d_mu Omega_nu - d_nu Omega_mu +
 ↪  [Omega_mu, Omega_nu]) = -(R/4) 1 with the curvature scalar R = -eta_44 (g55^2 + g66^2 + g77^2
 ↪  + gsum^2): the square of the Dirac operator differs from the wave operator of the spinor
 ↪  derivative by -R/4
[PASS] 71d-C8  (exact) transpose(Omega_mu).gamma1671[4] + gamma1671[4].Omega_mu = 0: the adjoint
 ↪  field psi^dagger gamma1671[4] is differentiated with -Omega_mu on its right, as the conserved
 ↪  current requires
[PASS] 71d-D1  (exact) in momentum space (d_mu -> -i k_mu) the Dirac operator of the weighted
 ↪  field at fixed x4 is gamma^mu k_mu - m with the coordinate matrices, and (gamma^mu k_mu +
 ↪  m)(gamma^nu k_nu - m) = (g^(mu nu) k_mu k_nu - m^2) 1 with g^(mu nu) k_mu k_nu = k_4^2 - k_0^2
 ↪  - k_1^2 - k_2^2 - k_3^2 - sum_t k_t^2/b_t^2: the instantaneous mass shell is k_4^2 = |p|^2 +
 ↪  sum_t k_t^2/b_t(x4)^2 + m^2. Verified for five rational parameter sets and four rational
 ↪  momenta each
[PASS] 71d-D2  (exact) the Hamiltonian of the weighted field, h(x4) = m beta1671 + sum_j p_j
 ↪  alpha1671^j + sum_t (k_t/b_t(x4)) alpha1671^t (beta1671 = gamma1671[4], alpha1671^A =
 ↪  gamma1671[4].gamma1671[A]), is real and symmetric for every momentum and every x4, and h^2 =
 ↪  (m^2 + |p|^2 + sum_t k_t^2/b_t^2) 1: the evolution in x4 is unitary and psihat^dagger psihat
 ↪  is conserved, but the energy of a mode with k_t != 0 grows as exp(rate_t x4)
[PASS] 71d-D3  (exact) dh/dx4 = sum_t rate_t (k_t/b_t) alpha1671^t is real symmetric and
 ↪  anticommutes with beta1671: the Hamiltonian changes with x4 only through the momenta along the
 ↪  deflating coordinates, which is what creates pairs
[PASS] 71d-D5  (exact) omega1671 commutes with every alpha1671^A and anticommutes with beta1671;
 ↪  hence the charge-conjugate field omega1671 psi^* solves the Dirac equation on the deflating
 ↪  spacetime whenever psi does (the scale factors are real)
[PASS] 71d-D6  (exact) with the chiral 16x8 matrices E^L, E^R of the main project, (E^L)^dagger h
 ↪  E^L = -i sum_j (p_j/e_j) tau1671^j, (E^R)^dagger h E^R = +i sum_j (p_j/e_j) tau1671^j and
 ↪  (E^L)^dagger h E^R = (E^R)^dagger h E^L = m, where p_j/e_j = (p_0, p_1, p_2, p_3, k_5/b_5,
 ↪  k_6/b_6, k_7/b_7): the equation of the weighted field is the pair (i d_4 + tau^j e_j^mu d_mu)
 ↪  phihat^L = m phihat^R, (i d_4 - tau^j e_j^mu d_mu) phihat^R = m phihat^L of 8-component
 ↪  equations
[PASS] 71d-D7  (exact) worked example: m = 3, p = 0, (k_5, k_6, k_7) = (2, 0, 0) at the value of
 ↪  x4 where b_5 = 1/2 (so k_5/b_5 = 4): h^2 = 25 1, and (5 + h)/10, (5 - h)/10 are complementary
 ↪  projectors of trace 8: eight states of energy +5 and eight of energy -5 at this x4; one
 ↪  e-folding of the deflation later the energy is sqrt(9 + 16 e^2) = 11.28
[PASS] ALL  (exact) every check above passed
```

A.2 Solutions mode by mode

The file `data/mode_checks.txt`, written by `deflation modes`:

```
PROJECT spacetimes, Revision_Deflation: checks of the solutions of the Dirac equations on the
 ↪  deflating spacetimes (71d-M..: pair creation on 7+1; 44d-M..: bounded solutions on 4+4)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation modes
7 of 7 checks passed

[PASS] 71d-M1  (numerical) pair creation by the deflation, one direction: for E_perp/g55 = 0.25,
 ↪  0.5, .., 3.0 the number of particles created per spin state in a mode that starts empty agrees
 ↪  with 1/(exp(2 pi E_perp/g55) + 1) to a relative deviation of at most 3.2e-6 (the numbers range
 ↪  from 0.17 to 6.5e-9); the eight spin states are created equally; the evolved states stay
 ↪  orthonormal to 2.0e-13
[PASS] 71d-M2  (numerical) equal rates g55 = g66 = g77 = 1.3 (constants 1, 0.7, 2), mode with m =
 ↪  1, p = (0, 0.1, 0.2, 0.3), (k_5, k_6, k_7) = (0.5, -0.8, 0.3): the created number per spin
 ↪  state is 5.70627486e-3 for all eight states, the law 1/(exp(2 pi E_perp/g55) + 1) gives
 ↪  5.70627482e-3 (relative deviation 6.7e-9): with equal rates the result depends neither on the
 ↪  direction of k_t nor on the constants
```

```
[PASS] 71d-M3  (numerical) unequal rates (g55 = 1 with g66 = 2, 1.2 or 0.5), mode with wavenumbers
 ↪  along x5 and x6: whenever the second physical wavenumber is still below exp(-3) E_perp at the
 ↪  moment at which the first reaches E_perp, the mean created number is the Fermi-Dirac law with
 ↪  the rate of the coordinate that becomes energetic first, to within 1.2 per cent; when the two
 ↪  become energetic together the eight spin states are created unequally (largest ratio of the
 ↪  largest to the smallest eigenvalue in the scan: 110) and no closed form is known
 ↪  (data/creation_anisotropic.csv)
[PASS] 44d-M1  (numerical) modes of the 4+4 field that depend on the deflating times (five cases:
 ↪  one, two or three non-zero k_t; equal and unequal rates and constants): beyond the turning
 ↪  value of x4 exactly eight of the sixteen solutions stay bounded; integrated backwards to early
 ↪  times they contain positive- and negative-energy parts of equal weight (relative difference of
 ↪  the two Gram matrices at most 2.0e-14): they are standing waves in x4, not travelling particles
[PASS] 44d-M2  (numerical) the covariant charge psihat^dagger W psihat of every bounded solution
 ↪  vanishes (ratio to psihat^dagger psihat at most 1.0e-14), as it must for a conserved quantity
 ↪  of a solution that dies out
[PASS] 44d-M3  (numerical) between the turning value and the late x4 every bounded solution falls
 ↪  by exp(-S) times a factor of order one, S = Int sqrt(k_d^2 - E_perp^2) dx4 = 30 being the WKB
 ↪  exponent (largest difference between the logarithm of the factor and S in the five cases:
 ↪  0.53). Where one rate governs (cases 1 to 4) the eight solutions fall by the same factor
 ↪  (spread of the logarithms at most 1.1e-14) and its logarithm is S + (1/2) ln 2 - (1/2) ln(1 +
 ↪  exp(-2 pi E_perp/rate_t)), as the exact solution requires (largest deviation 3.6e-4); with
 ↪  three unequal rates (case 5) the factor depends on the spin state (spread 0.40). The
 ↪  dependence on a deflating time dies out as exp(-S), and S grows as exp(rate_t x4)
[PASS] 44d-M4  (numerical) where one rate governs (one non-zero k_t, or equal rates) the computed
 ↪  |psihat|^2 equals the exact form 2 u_d |K_(1/2 + i E_perp/rate_t)(u_d)|^2 (modified Bessel
 ↪  function of the order 1/2 + i E_perp/rate_t, u_d = k_d/rate_t) over the range 0.02 < u_d < 12
 ↪  with a largest relative deviation of 3.6e-12
[PASS] ALL  (numerical) every check above passed
```

A.3 Symmetries

The file `data/symmetry_checks.txt`, written by `deflation symmetry`. The checks `44d-V..`, `71d-V..` and `def-S4` belong to the supplement RS1 and are discussed there.

```
PROJECT spacetimes, Revision_Deflation: checks of the symmetries on the deflating spacetimes
 ↪  (isometries, internal symmetry of the fields that do not depend on the extra coordinates, zero
 ↪  modes on a vortex line, reduction of the couplings)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation symmetry
18 of 18 checks passed

[PASS] 44d-S1  (exact) of the 28 rotations and boosts of the flat 4+4 spacetime (one for each pair
 ↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
 ↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
 ↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
 ↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
 ↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
 ↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
 ↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 44d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
 ↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
 ↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
 ↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
 ↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
 ↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
 ↪  symmetries)
```

```
[PASS] 71d-S1  (exact) of the 28 rotations and boosts of the flat 7+1 spacetime (one for each pair
↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 71d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
↪  symmetries)
[PASS] def-S3  (numerical) the internal symmetry of the fields that do not depend on the extra
↪  coordinates, computed again for both spacetimes (24 items of data/reduced_symmetry.csv, all
↪  as expected): the commutant of the five matrices has dimension 8 and is u(2) + u(2) = su(2)_A
↪  + su(2)_B + u(1)_F + u(1)_E; the eight one-particle states are (2,1) + (1,2); the four vector
↪  pair operators are the (2,2) with fermion number 2, and their invariant quadratic form is the
↪  frame metric of the axes x0, x5, x6, x7. The deflation does not change this symmetry, because
↪  the weighted field of this sector obeys the flat equations (checks 44d-D.., 71d-D.. of the
↪  command matrices)
[PASS] 44d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
↪  one and anticommute with one another
[PASS] 44d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
↪  square one
[PASS] 44d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 44d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
↪  the generator P of U(1)_E
[PASS] 44d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 44d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] 71d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
↪  one and anticommute with one another
[PASS] 71d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
↪  square one
[PASS] 71d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 71d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
↪  the generator P of U(1)_E
[PASS] 71d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 71d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] def-S4  (exact) reduction to fields that do not depend on x5, x6, x7: sqrt|g| times the
↪  Yukawa term and the potential of the Vorton-Scalar, written with the weighted fields psihat =
↪  sqrt(Vd) psi and Phihat = sqrt(Vd) Phi_VS, are the same expressions with the couplings
↪  y_VS/sqrt(Vd), lambda_VS/Vd and the vacuum value v_VS sqrt(Vd); the products y_VS v_VS and
↪  lambda_VS v_VS^2, hence the fermion mass and the mass of the radial mode, do not change
↪  (rational sample values, exact). The exponents of data/coupling_exponents.csv follow, with Vd
↪  proportional to exp(-gsum x4)
```

```
[PASS] ALL  (exact) every check above passed
```

A.4 The matrices of the frame, verified again

The file `data/flat_matrix_checks.txt`, written by `deflation matrices` through the routines of the main project. Its statements use the symbols of the main registry, and its section references are to the document `dirac matrices`.

```
PROJECT spacetimes, Revision_Deflation: the checks of the real 16x16 Dirac matrices of the main
 ↪  project, executed again (these matrices are the matrices of the orthonormal frame of the
 ↪  deflating spacetimes); identifiers 44-.. refer to gamma1644, 71-.. to gamma1671
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation matrices
106 of 106 checks passed

[PASS] 44-R1  (exact) every entry of gamma1644[A], A = 0..7, is one of the integers -1, 0, +1;
 ↪  hence the eight matrices are real
[PASS] 44-R2  (exact) every gamma1644[A] is a signed permutation matrix (one entry +1 or -1 in
 ↪  each row and in each column)
[PASS] 44-C1  (exact) gamma1644[A].gamma1644[B] + gamma1644[B].gamma1644[A] = 2 eta1644^(AB) 1
 ↪  for all 64 pairs (A, B), with eta1644 = diag[1, 1, 1, 1, -1, -1, -1, -1]
[PASS] 44-C2  (exact) transpose(gamma1644[A]) = eta1644^(AA) gamma1644[A]: matrices with square
 ↪  +1 are symmetric, matrices with square -1 are antisymmetric; hence transpose(gamma1644[A]) =
 ↪  inverse(gamma1644[A])
[PASS] 44-C3  (exact) the 256 ordered products gamma1644[A1]...gamma1644[Ak] (A1 < ... < Ak, k =
 ↪  0..8) are linearly independent: they span all real 16x16 matrices, so the representation is
 ↪  irreducible over the real numbers
[PASS] 44-C4  (exact) every one of the 255 non-empty products has trace 0, and the square of every
 ↪  product is +1 or -1
[PASS] 44-L1  (exact) Sigma1644^(AB) := (1/4)[gamma1644[A], gamma1644[B]] equals (1/2)
 ↪  gamma1644[A].gamma1644[B] for A != B, and Sigma1644^(AB) = -Sigma1644^(BA)
[PASS] 44-L2  (exact) the 28 matrices Sigma1644^(AB), A < B, are linearly independent
[PASS] 44-L3  (exact) [Sigma1644^(AB), gamma1644[C]] = eta1644^(BC) gamma1644[A] - eta1644^(AC)
 ↪  gamma1644[B] for all A, B, C: the Sigma1644^(AB) rotate the eight matrices as the components
 ↪  of a vector
[PASS] 44-L4  (exact) [Sigma1644^(AB), Sigma1644^(CD)] = eta1644^(BC) Sigma1644^(AD) -
 ↪  eta1644^(AC) Sigma1644^(BD) - eta1644^(BD) Sigma1644^(AC) + eta1644^(AD) Sigma1644^(BC) for
 ↪  all 4096 index combinations: the commutation relations of the Lie algebra so(p,q) of the
 ↪  metric eta1644
[PASS] 44-L5  (exact) the structure constants are integers and the Killing form is diagonal in the
 ↪  basis Sigma1644^(AB): B(Sigma1644^(AB), Sigma1644^(AB)) = -12 eta1644^(AA) eta1644^(BB); it
 ↪  has 16 positive and 12 negative eigenvalues (no zero eigenvalue: the algebra is semisimple)
[PASS] 44-L6  (exact) 12 generators are antisymmetric matrices (compact: they generate rotations)
 ↪  and 16 are symmetric (non-compact: they generate boosts); these are the numbers of the real
 ↪  form so(4,4), which has 12 compact and 16 non-compact generators
[PASS] 44-G1  (exact) each gamma1644[A] is invertible with inverse eta1644^(AA) gamma1644[A], and
 ↪  -gamma1644[A].gamma1644[B].inverse(gamma1644[A]) = gamma1644[B] for B != A, = -gamma1644[A]
 ↪  for B = A: gamma1644[A] acts on the vector index as the reflection that reverses the axis xA
[PASS] 44-G2  (numerical) exp(theta Sigma1644^(AB)) = cos(theta/2) 1 + sin(theta/2)
 ↪  gamma1644[A].gamma1644[B] when (gamma1644[A].gamma1644[B])^2 = -1 (rotation), and =
 ↪  cosh(theta/2) 1 + sinh(theta/2) gamma1644[A].gamma1644[B] when (gamma1644[A].gamma1644[B])^2
 ↪  = +1 (boost), for the 28 planes and random theta (largest deviation 2.3e-14)
[PASS] 44-G3  (numerical) for 20 random elements S = exp(sum_(A<B) theta_AB Sigma1644^(AB))
 ↪  (theta_AB random real numbers):  S.gamma1644[C].inverse(S) = sum_D Lambda^C_D gamma1644[D]
 ↪  with a real 8x8 matrix Lambda that preserves eta1644 (Lambda.eta1644.transpose(Lambda) =
 ↪  eta1644) and has determinant +1 (largest deviations 2.0e-13, 2.2e-15, 2.7e-15)
[PASS] 44-N1  (exact) the eight matrices rebuilt from the definitions equal, entry by entry, the
 ↪  eight matrices T16[0..7] printed by Mathematica in 4+4.nb (cell 314, Out[211])
[PASS] 44-N2  (exact) the rebuilt volume element equals T16[8] printed in 4+4.nb (cell 136,
 ↪  Out[99])
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[PASS] 44-N3  (exact) the metric printed in 4+4.nb (cell 13, Out[9]) is eta4488 =
↪  diag(1,1,1,1,-1,-1,-1,-1) (PROJECT name: eta1644)
[PASS] 44-B1  (exact) the 8x8 matrix sigma (= sigma88) of 4+4.nb, PROJECT name sigma88: sigma88 =
↪  tau[1].tau[2].tau[3] = tau[4].tau[5].tau[6].tau[7], sigma88.sigma88 = 1,
↪  tau[1].tau[2]...tau[7] = 1, and tau[7] = diag(-1_4, +1_4) (4+4.nb Out[72], Out[89], Out[87],
↪  Out[63])
[PASS] 44-B2  (exact) tau[A].taubar[B] + tau[B].taubar[A] = 2 eta1644^(AB) 1 and taubar[A].tau[B]
↪  + taubar[B].tau[A] = 2 eta1644^(AB) 1 (8x8), A, B = 0..7 (4+4.nb Out[77])
[PASS] 44-B3  (exact) sigma88.taubar[A] = transpose(sigma88.tau[A]) for A = 0..7 (4+4.nb Out[76])
[PASS] 44-B4  (exact) T16[A] = [[0, taubar[A]], [tau[A], 0]] in 8x8 blocks, with tau[0] =
↪  taubar[0] = 1.  PROJECT names of these objects of 4+4.nb: gamma1644[A] := T16[A], tau1644[A]
↪  := tau[A], taubar1644[A] := taubar[A]
[PASS] 44-V1  (exact) omega1644 := gamma1644[0]...gamma1644[7] (= T16[8]) = diag(-1_8, +1_8):
↪  real, diagonal, symmetric, with square +1
[PASS] 44-V2  (exact) omega1644 anticommutes with every gamma1644[A] and commutes with every
↪  Sigma1644^(AB)
[PASS] 44-P1  (exact) PL1644 := (1 - omega1644)/2 = diag(1_8, 0_8) and PR1644 := (1 + omega1644)/2
↪  = diag(0_8, 1_8) are real matrices with integer entries
[PASS] 44-P2  (exact) PL1644 + PR1644 = 1, PL1644.PL1644 = PL1644, PR1644.PR1644 = PR1644,
↪  PL1644.PR1644 = PR1644.PL1644 = 0, trace PL1644 = trace PR1644 = 8 (4+4.nb Out[114])
[PASS] 44-P3  (exact) the rebuilt projectors PL1644 and PR1644 equal the matrices P_L and P_R
↪  printed in 4+4.nb (cells 163 and 165, Out[112] and Out[113])
[PASS] 44-P4  (exact) gamma1644[A].PL1644 = PR1644.gamma1644[A] and gamma1644[A].PR1644 =
↪  PL1644.gamma1644[A]: every Dirac matrix maps left-handed spinors to right-handed ones and
↪  conversely
[PASS] 44-P5  (exact) PL1644 and PR1644 commute with every Sigma1644^(AB): chirality is invariant
↪  under Spin(4,4)
[PASS] 44-S1  (exact) Cm1644 := gamma1644[0].gamma1644[1].gamma1644[2].gamma1644[3] (the matrix
↪  sigma16 of 4+4.nb) = diag(-sigma88, sigma88), symmetric, Cm1644.Cm1644 = 1 (4+4.nb Out[106])
[PASS] 44-S2  (exact) Cp1644 := gamma1644[4].gamma1644[5].gamma1644[6].gamma1644[7] (the matrix
↪  S44 of 7+1.nb) = diag(sigma88, sigma88), symmetric, Cp1644.Cp1644 = 1, and omega1644 =
↪  Cm1644.Cp1644
[PASS] 44-S3  (exact) Cm1644.gamma1644[A].Cm1644 = -transpose(gamma1644[A]) and
↪  Cp1644.gamma1644[A].Cp1644 = +transpose(gamma1644[A]) for A = 0..7
[PASS] 44-S4  (exact) Cm1644.gamma1644[A] is antisymmetric for A = 0..7 (4+4.nb Out[105]) and
↪  Cp1644.gamma1644[A] is symmetric
[PASS] 44-S5  (exact) gamma1644[4] (the matrix JT4 = T16[4] of 4+4.nb) is a complex structure:
↪  gamma1644[4].gamma1644[4] = -1, and transpose(gamma1644[4]).Cm1644.gamma1644[4] = Cm1644
[PASS] 71-R1  (exact) every entry of gamma1671[A], A = 0..7, is one of the integers -1, 0, +1;
↪  hence the eight matrices are real
[PASS] 71-R2  (exact) every gamma1671[A] is a signed permutation matrix (one entry +1 or -1 in
↪  each row and in each column)
[PASS] 71-C1  (exact) gamma1671[A].gamma1671[B] + gamma1671[B].gamma1671[A] = 2 eta1671^(AB) 1
↪  for all 64 pairs (A, B), with eta1671 = diag[-1, -1, -1, -1, 1, -1, -1, -1]
[PASS] 71-C2  (exact) transpose(gamma1671[A]) = eta1671^(AA) gamma1671[A]: matrices with square
↪  +1 are symmetric, matrices with square -1 are antisymmetric; hence transpose(gamma1671[A]) =
↪  inverse(gamma1671[A])
[PASS] 71-C3  (exact) the 256 ordered products gamma1671[A1]...gamma1671[Ak] (A1 < ... < Ak, k =
↪  0..8) are linearly independent: they span all real 16x16 matrices, so the representation is
↪  irreducible over the real numbers
[PASS] 71-C4  (exact) every one of the 255 non-empty products has trace 0, and the square of every
↪  product is +1 or -1
[PASS] 71-L1  (exact) Sigma1671^(AB) := (1/4)[gamma1671[A], gamma1671[B]] equals (1/2)
↪  gamma1671[A].gamma1671[B] for A != B, and Sigma1671^(AB) = -Sigma1671^(BA)
[PASS] 71-L2  (exact) the 28 matrices Sigma1671^(AB), A < B, are linearly independent
[PASS] 71-L3  (exact) [Sigma1671^(AB), gamma1671[C]] = eta1671^(BC) gamma1671[A] - eta1671^(AC)
↪  gamma1671[B] for all A, B, C: the Sigma1671^(AB) rotate the eight matrices as the components
↪  of a vector
[PASS] 71-L4  (exact) [Sigma1671^(AB), Sigma1671^(CD)] = eta1671^(BC) Sigma1671^(AD) -
↪  eta1671^(AC) Sigma1671^(BD) - eta1671^(BD) Sigma1671^(AC) + eta1671^(AD) Sigma1671^(BC) for
↪  all 4096 index combinations: the commutation relations of the Lie algebra so(p,q) of the
↪  metric eta1671
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[PASS] 71-L5  (exact) the structure constants are integers and the Killing form is diagonal in the
 ↪  basis Sigma1671^(AB): B(Sigma1671^(AB), Sigma1671^(AB)) = -12 eta1671^(AA) eta1671^(BB); it
 ↪  has 7 positive and 21 negative eigenvalues (no zero eigenvalue: the algebra is semisimple)
[PASS] 71-L6  (exact) 21 generators are antisymmetric matrices (compact: they generate rotations)
 ↪  and 7 are symmetric (non-compact: they generate boosts); these are the numbers of the real
 ↪  form so(7,1), which has 21 compact and 7 non-compact generators
[PASS] 71-G1  (exact) each gamma1671[A] is invertible with inverse eta1671^(AA) gamma1671[A], and
 ↪  -gamma1671[A].gamma1671[B].inverse(gamma1671[A]) = gamma1671[B] for B != A, = -gamma1671[A]
 ↪  for B = A: gamma1671[A] acts on the vector index as the reflection that reverses the axis xA
[PASS] 71-G2  (numerical) exp(theta Sigma1671^(AB)) = cos(theta/2) 1 + sin(theta/2)
 ↪  gamma1671[A].gamma1671[B] when (gamma1671[A].gamma1671[B])^2 = -1 (rotation), and =
 ↪  cosh(theta/2) 1 + sinh(theta/2) gamma1671[A].gamma1671[B] when (gamma1671[A].gamma1671[B])^2
 ↪  = +1 (boost), for the 28 planes and random theta (largest deviation 7.1e-15)
[PASS] 71-G3  (numerical) for 20 random elements S = exp(sum_(A<B) theta_AB Sigma1671^(AB))
 ↪  (theta_AB random real numbers):  S.gamma1671[C].inverse(S) = sum_D Lambda^C_D gamma1671[D]
 ↪  with a real 8x8 matrix Lambda that preserves eta1671 (Lambda.eta1671.transpose(Lambda) =
 ↪  eta1671) and has determinant +1 (largest deviations 5.0e-14, 2.0e-15, 2.0e-15)
[PASS] 71-N1  (exact) the eight matrices rebuilt from the definitions equal, entry by entry, the
 ↪  eight matrices Gamma17[0..7] printed by Mathematica in 7+1.nb (cell 204, Out[178])
[PASS] 71-N2  (exact) PROJECT matrices: gamma1671[A] = Gamma17up[swap04(A)] with swap04 = (0 4)
 ↪  and Gamma17up[A] = eta17^(AA) Gamma17[A] (7+1.nb section 5.3); explicitly gamma1671[4] =
 ↪  Gamma17[0] (time-like), gamma1671[0] = -Gamma17[4], gamma1671[j] = -Gamma17[j] for j = 1, 2,
 ↪  3, 5, 6, 7.  (gamma1671 is NOT the complex set Gamma71 of 7+1.nb section 4.3, and eta1671 is
 ↪  NOT the matrix eta71 of 7+1.nb)
[PASS] 71-N3  (exact) in the labelling of 7+1.nb: Gamma17[A].Gamma17[B] + Gamma17[B].Gamma17[A] =
 ↪  2 eta17^(AB) 1 with eta17 = diag(+1,-1,-1,-1,-1,-1,-1,-1) (7+1.nb Out[173])
[PASS] 71-B1  (exact) tauO[A] = tau[A] (A = 1, 2, 3), tauO[A] = tau[A].sigma (A = 4..7): real,
 ↪  antisymmetric, tauO[A].tauO[A] = -1, mutually anticommuting, tauO[1]...tauO[7] = 1 (7+1.nb
 ↪  section 3.1)
[PASS] 71-B2  (exact) block form: Gamma17[0] = [[0, 1], [1, 0]] and Gamma17[i] = [[tauO[i], 0],
 ↪  [0, tauObar[i]]] with tauObar[i] = transpose(tauO[i]) = -tauO[i], i = 1..7 (7+1.nb Out[175])
[PASS] 71-B3  (exact) built from the 4+4 matrices only: Gamma17[0] = T16[0], Gamma17[i] =
 ↪  T16[0].T16[i] (i = 1, 2, 3), Gamma17[k] = T16[0].T16[k].S44 (k = 4..7) (7+1.nb Out[176])
[PASS] 71-B4  (exact) in the labels of this project: gamma1671[4] = gamma1644[0]; gamma1671[j] =
 ↪  -gamma1644[0].gamma1644[j] (j = 1, 2, 3); gamma1671[0] =
 ↪  +gamma1644[0].gamma1644[5].gamma1644[6].gamma1644[7]; gamma1671[5] =
 ↪  -gamma1644[0].gamma1644[4].gamma1644[6].gamma1644[7]; gamma1671[6] =
 ↪  +gamma1644[0].gamma1644[4].gamma1644[5].gamma1644[7]; gamma1671[7] =
 ↪  -gamma1644[0].gamma1644[4].gamma1644[5].gamma1644[6].  The time-like matrix of 7+1 IS the
 ↪  matrix of the hidden axis of 4+4; the other seven are products of two or of four of the 4+4
 ↪  matrices and therefore not linear combinations of them
[PASS] 71-V1  (exact) omega1671 := gamma1671[0]...gamma1671[7] = Gamma17[8] = [[0, -1], [1, 0]] in
 ↪  8x8 blocks (the omega17 of 7+1.nb: the seven sign changes of Gamma17up and the odd permutation
 ↪  swap04 compensate): real, antisymmetric, with square -1
[PASS] 71-V2  (exact) omega1671 anticommutes with every gamma1671[A] and commutes with every
 ↪  Sigma1671^(AB)
[PASS] 71-V3  (exact) the linear system X.gamma1671[A] + gamma1671[A].X = 0 (A = 0..7) for a real
 ↪  16x16 matrix X has rank 255, and omega1671 solves it (71-V2): the real matrices that
 ↪  anticommute with all eight gamma1671[A] are exactly the multiples of omega1671; since
 ↪  omega1671.omega1671 = -1, no real matrix with square +1 anticommutes with all of them: there
 ↪  is NO real chirality matrix in 7+1
[PASS] 71-P1  (exact) chi1671 := i omega1671 = i Gamma17[8] (= chi17 of 7+1.nb) is Hermitian,
 ↪  chi1671.chi1671 = 1, anticommutes with every gamma1671[A]
[PASS] 71-P2  (exact) PL1671 := (1 - chi1671)/2 and PR1671 := (1 + chi1671)/2: PL1671 + PR1671 =
 ↪  1, PL1671.PL1671 = PL1671, PR1671.PR1671 = PR1671, PL1671.PR1671 = 0, Hermitian, trace 8 each;
 ↪  they are complex (not real) matrices
[PASS] 71-P3  (exact) gamma1671[A].PL1671 = PR1671.gamma1671[A] for A = 0..7, and PL1671, PR1671
 ↪  commute with every Sigma1671^(AB)
[PASS] 71-P4  (exact) in 8x8 blocks chi1671 = [[0, -i], [i, 0]], 2 PL1671 = [[1, i], [-i, 1]] and
 ↪  2 PR1671 = [[1, -i], [i, 1]]; these are chi17, PL17 and PR17 of 7+1.nb (sections 4.6 and 7.10)
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[PASS] 71-P5  (exact) with s := 1 - i Gamma17[0] (s.s^dagger = 2, so Uc1671 := s/sqrt(2) is
 ↪  unitary): s.chi1671 = -T16[8].s, i.e. Uc1671.chi1671.inverse(Uc1671) = -omega1644.  Hence
 ↪  Uc1671.PL1671.inverse(Uc1671) = PR1644 and Uc1671.PR1671.inverse(Uc1671) = PL1644: the
 ↪  complex chiral projectors of 7+1 are unitarily equivalent to the real projectors of 4+4.nb,
 ↪  with the names L and R exchanged.  (7+1.nb defines chi17 = +i Gamma17[8] for its real matrices
 ↪  and chi71 = -i Gamma71[8] for its complex matrices Gamma71; the PROJECT matrix chi1671 equals
 ↪  chi17, NOT chi71)
[PASS] 71-S1  (exact) Dirac adjoint: transpose(gamma1671[A]) =
 ↪  gamma1671[4].gamma1671[A].gamma1671[4] for A = 0..7 (gamma1671[4] is the time-like matrix; it
 ↪  is symmetric and squares to +1)
[PASS] 44-K1  (exact) 4+4: the matrices Cm1644.gamma1644[A1]...gamma1644[Ar] (A1 < ... < Ar) of
 ↪  rank r = 0..8 are all symmetric (+) or all antisymmetric (-): signs [1, -1, -1, 1, 1, -1, -1,
 ↪  1, 1]; the matrices Cp1644.gamma1644[A1]...gamma1644[Ar]: signs [1, 1, -1, -1, 1, 1, -1, -1,
 ↪  1]; in both cases exactly 120 of the 256 matrices are antisymmetric
[PASS] 44-K2  (exact) 4+4: the two Lorentz-scalar pair matrices Cm1644 (rank 0) and
 ↪  Cm1644.omega1644 (rank 8; equal to Cp1644 up to sign) are SYMMETRIC: the Lorentz-scalar
 ↪  bilinears dirac1644^T Cm1644 dirac1644 and dirac1644^T Cp1644 dirac1644 of one anticommuting
 ↪  16-component field vanish identically
[PASS] 71-K0  (exact) 7+1: Cp1671 := gamma1671[4] satisfies Cp1671.gamma1671[A] =
 ↪  +transpose(gamma1671[A]).Cp1671 and Cm1671 := gamma1671[4].omega1671 satisfies
 ↪  Cm1671.gamma1671[A] = -transpose(gamma1671[A]).Cm1671; both are symmetric matrices
[PASS] 71-K1  (exact) 7+1: signs (+ symmetric, - antisymmetric) of the matrices
 ↪  Cm1671.gamma1671[A1]...gamma1671[Ar] (A1 < ... < Ar) by rank r = 0..8: [1, -1, -1, 1, 1, -1,
 ↪  -1, 1, 1]; of Cp1671.gamma1671[A1]...gamma1671[Ar]: [1, 1, -1, -1, 1, 1, -1, -1, 1]; exactly
 ↪  120 of the 256 matrices are antisymmetric in both cases
[PASS] 71-K2  (exact) 7+1: the two Lorentz-scalar pair matrices Cm1671 (rank 0) and
 ↪  Cm1671.omega1671 = -Cp1671 (rank 8) are SYMMETRIC: the Lorentz-scalar bilinears dirac1671^T
 ↪  Cm1671 dirac1671 and dirac1671^T Cp1671 dirac1671 of one anticommuting 16-component field
 ↪  vanish identically
[PASS] 44-K3  (exact) 4+4: for all A != B in 1..7, transpose(Sigma1644^(AB)).Cm1644.gamma1644[0] +
 ↪  Cm1644.gamma1644[0].Sigma1644^(AB) = 0, while the same expression with A = 0 is not zero: the
 ↪  bilinear DeltaV1644^0 = dirac1644^T Cm1644 gamma1644^0 dirac1644 is INVARIANT under the 21
 ↪  generators of the pseudo-rotations of the seven coordinates x1..x7 (the group Spin(3,4), which
 ↪  contains the Lorentz group of x1 x2 x3 x4), and only the 7 generators that involve the hidden
 ↪  axis x0 change it
[PASS] 71-K3  (exact) 7+1: for all A != B in 1..7, transpose(Sigma1671^(AB)).Cm1671.gamma1671[0] +
 ↪  Cm1671.gamma1671[0].Sigma1671^(AB) = 0, while the same expression with A = 0 is not zero: the
 ↪  bilinear DeltaV1671^0 = dirac1671^T Cm1671 gamma1671^0 dirac1671 is INVARIANT under the 21
 ↪  generators of the Lorentz group of the seven coordinates x1..x7 (the group Spin(6,1), which
 ↪  contains the Lorentz group of x1 x2 x3 x4), and only the 7 generators that involve the hidden
 ↪  axis x0 change it
[PASS] 44-D1  (exact) (gamma1644^A k_A)^2 = (eta1644^(AB) k_A k_B) 1 for 40 random integer vectors
 ↪  k: the Dirac operator gamma1644^A d_A squares to the wave operator eta1644^(AB) d_A d_B = d0^2
 ↪  + d1^2 + d2^2 + d3^2 - d4^2 - d5^2 - d6^2 - d7^2
[PASS] 44-D2  (exact) (i gamma1644.k + m1644)(i gamma1644.k - m1644) = -(k.k + m1644^2) 1 for 40
 ↪  random integer (k, m1644): every component of a solution of (gamma1644^A d_A - m1644)
 ↪  dirac1644 = 0 obeys the Klein-Gordon equation (eta1644^(AB) d_A d_B - m1644^2) dirac1644 = 0
[PASS] 44-D3  (exact) worked example k = (0,2,2,4; 5,0,0,0), m1644 = 1 (on the mass shell:
 ↪  k4^2+k5^2+k6^2+k7^2 = k0^2+k1^2+k2^2+k3^2 + m1644^2): Pi1644 := (i gamma1644.k + m1644)/(2
 ↪  m1644) satisfies Pi1644.Pi1644 = Pi1644, trace Pi1644 = 8 and (i gamma1644.k - m1644).Pi1644 =
 ↪  0, so the plane-wave solutions u exp(i k.x) form an 8-dimensional space, u = Pi1644 w
[PASS] 44-D4  (exact) PL1644.gamma1644[A].PR1644 = [[0, taubar1644[A]], [0, 0]],
 ↪  PR1644.gamma1644[A].PL1644 = [[0, 0], [tau1644[A], 0]], PL1644.gamma1644[A].PL1644 =
 ↪  PR1644.gamma1644[A].PR1644 = 0: with dirac1644 = (psiL1644, psiR1644) the 16-component
 ↪  equation is the pair  taubar1644^A d_A psiR1644 = m1644 psiL1644,  tau1644^A d_A psiL1644 =
 ↪  m1644 psiR1644  of 8-component equations
[PASS] 44-D5  (exact) (tau1644^A k_A)(taubar1644^B k_B) = (taubar1644^A k_A)(tau1644^B k_B) =
 ↪  (eta1644^(AB) k_A k_B) 1_8 for 40 random integer k: each 8-component field obeys the
 ↪  Klein-Gordon equation
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[PASS] 44-D6  (exact) gamma1644[4].gamma1644[a] (a = 0..3) are real symmetric with square +1 (and
 ↪  so are alpha1644^a := -gamma1644[4].gamma1644[a], the matrices of the Hamiltonian h1644 =
 ↪  alpha1644^a k_a + beta1644 m1644, beta1644 := -i gamma1644[4]); gamma1644[4].gamma1644[t] (t
 ↪  = 5, 6, 7) are real antisymmetric with square -1; gamma1644[4] is real antisymmetric with
 ↪  square -1
[PASS] 44-D7  (exact) x4-evolution d_4 dirac1644 = K1644 dirac1644 with K1644 = -m1644 gamma1644^4
 ↪  + i k_a gamma1644^4 gamma1644^a + i kappa_t gamma1644^4 gamma1644^t (a = 0..3 space, t = 5, 6,
 ↪  7 extra times): K1644.K1644 = -(m1644^2 + |k|^2 - |kappa|^2) 1 for 40 random integer (k,
 ↪  kappa, m1644): the frequencies in x4 are +-sqrt(m1644^2 + |k|^2 - |kappa|^2)
[PASS] 44-D8  (exact) K1644 + K1644^dagger = 2 i kappa_t gamma1644^4 gamma1644^t: K1644 is
 ↪  anti-Hermitian (so that h1644 := i K1644 is Hermitian and the x4-evolution preserves
 ↪  dirac1644^dagger dirac1644) if and only if kappa = 0, i.e. if the field does not depend on the
 ↪  extra times x5, x6, x7
[PASS] 44-D9  (exact) example of the instability: m1644 = 3, k = 0, kappa = (5, 0, 0) gives
 ↪  K1644.K1644 = +16, (K1644 - 4)(K1644 + 4) = 0 with trace K1644 = 0: K1644 has the real
 ↪  eigenvalues +4 and -4, eight times each, so eight solutions grow like exp(+4 x4)
[PASS] 44-D10  (exact) W1644 := i Cm1644.gamma1644[4] is Hermitian with W1644.W1644 = 1 and trace
 ↪  0 (eigenvalues +1 and -1, eight times each); W1644 commutes with gamma1644[4] and with
 ↪  gamma1644[4].gamma1644[a] (a = 0..3), hence with K1644 at kappa = 0, and anticommutes with
 ↪  gamma1644[4].gamma1644[t] (t = 5, 6, 7)
[PASS] 44-D11  (exact) the density of the conserved current of the Spin(4,4)-invariant Lagrangian,
 ↪  j1644^4 = dirac1644^dagger (i Cm1644.gamma1644[4]) dirac1644 = dirac1644^dagger W1644
 ↪  dirac1644, is NOT positive: W1644 has eight eigenvalues -1; the positive density
 ↪  dirac1644^dagger dirac1644 is conserved only for kappa = 0 (44-D8)
[PASS] 71-D1  (exact) (gamma1671^A k_A)^2 = (eta1671^(AB) k_A k_B) 1 = (k4^2 - k0^2 - k1^2 - k2^2
 ↪  - k3^2 - k5^2 - k6^2 - k7^2) 1 for 40 random integer vectors k
[PASS] 71-D2  (exact) (gamma1671.k + m1671)(gamma1671.k - m1671) = (k.k - m1671^2) 1 for 40 random
 ↪  integer (k, m1671): with dirac1671 = u exp(-i k_A x^A) the operator i gamma1671^A d_A - m1671
 ↪  becomes gamma1671.k - m1671, and every component of a solution obeys the Klein-Gordon equation
 ↪  (d4^2 - d0^2 - d1^2 - d2^2 - d3^2 - d5^2 - d6^2 - d7^2 + m1671^2) dirac1671 = 0
[PASS] 71-D3  (exact) worked example E = 5, momentum p = (p1, p2, p3) = (2, 2, 4), m1671 = 1 (on
 ↪  the mass shell, 25 = 24 + 1): Pi1671 := (gamma1671.k + m1671)/(2 m1671) satisfies
 ↪  Pi1671.Pi1671 = Pi1671, trace Pi1671 = 8, (gamma1671.k - m1671).Pi1671 = 0: eight independent
 ↪  positive-energy plane waves
[PASS] 71-D4  (exact) beta1671 := gamma1671[4] and alpha1671^j := gamma1671[4].gamma1671[j] (j =
 ↪  0, 1, 2, 3, 5, 6, 7) are real symmetric; alpha1671^i.alpha1671^j + alpha1671^j.alpha1671^i = 2
 ↪  delta^(ij), alpha1671^j.beta1671 + beta1671.alpha1671^j = 0, beta1671.beta1671 = 1: the
 ↪  Hamiltonian h1671 = -i alpha1671^j d_j + beta1671 m1671 is Hermitian
[PASS] 71-D5  (exact) h1671(p) = p^j alpha1671^j + m1671 beta1671 is a real symmetric matrix with
 ↪  h1671(p).h1671(p) = (|p|^2 + m1671^2) 1 for 40 random integer (p, m1671): the energies are +E
 ↪  and -E, E = sqrt(|p|^2 + m1671^2), eight times each
[PASS] 71-D6  (exact) tau1671^j := tauO[swap04(j)] (swap04 = (0 4)): tau1671^4 = 1_8; for the seven
 ↪  space indices tau1671^j is real, antisymmetric, tau1671^i.tau1671^j + tau1671^j.tau1671^i =
 ↪  -2 delta^(ij); in 8x8 blocks beta1671 = [[0, 1], [1, 0]] and alpha1671^j = [[0, tau1671^j],
 ↪  [-tau1671^j, 0]] (beta1671 = betaD and alpha1671^j = alphaD[swap04(j)] of 7+1.nb section 7.1)
[PASS] 71-D7  (exact) chiral form: with EL1671 = (1; -i)/sqrt(2), ER1671 = (-i; 1)/sqrt(2) (16x8,
 ↪  in 8x8 blocks; chi1671.EL1671 = -EL1671, chi1671.ER1671 = +ER1671, orthonormal):
 ↪  EL1671^dagger.h1671(p).EL1671 = -i p^j tau1671^j, ER1671^dagger.h1671(p).ER1671 = +i p^j
 ↪  tau1671^j, EL1671^dagger.h1671(p).ER1671 = ER1671^dagger.h1671(p).EL1671 = m1671 (20 random
 ↪  integer (p, m1671)); hence (i d_4 + tau1671^j d_j) phiL1671 = m1671 phiR1671 and (i d_4 -
 ↪  tau1671^j d_j) phiR1671 = m1671 phiL1671 for phiL1671 = EL1671^dagger dirac1671, phiR1671 =
 ↪  ER1671^dagger dirac1671 (EL1671, ER1671, phiL1671, phiR1671 are the objects EL, ER, phiL, phiR
 ↪  of 7+1.nb sections 7.10 and 7.11, in the PROJECT labelling of the coordinates)
[PASS] 71-D8  (exact) worked example E = 5, p = (2, 2, 4), m1671 = 1: with M1671 = m1671 + p^j
 ↪  tau1671^j (M1671.transpose(M1671) = E^2 1_8), the columns of Upos1671 := (E 1;
 ↪  transpose(M1671)) satisfy h1671(p).Upos1671 = +E Upos1671 and those of Vneg1671 := (E 1;
 ↪  -M1671) satisfy h1671(-p).Vneg1671 = -E Vneg1671; transpose(Upos1671).Upos1671 =
 ↪  transpose(Vneg1671).Vneg1671 = 2 E^2 1_8: the spinors u_s = Upos1671 e_s/sqrt(E), v_s =
 ↪  Vneg1671 e_s/sqrt(E) are REAL and normalized to 2E
```

[PASS] 71-D9 (exact) charge conjugation dirac1671_c := omega1671 conj(dirac1671): omega1671 is real and omega1671.gamma1671[A] = -gamma1671[A].omega1671, so (i gamma1671^A d_A - m1671) dirac1671_c = omega1671 conj((i gamma1671^A d_A - m1671) dirac1671); omega1671.omega1671 = -1 gives (dirac1671_c)_c = -dirac1671: a 16-component Majorana condition dirac1671_c = dirac1671 has only the solution dirac1671 = 0
[PASS] 71-Q1 (numerical) 7+1, one particle at rest: the 8 spin states form the spinor of SO(7) (Casimir 21/4) and split under SO(3) x SU(2)_A x SU(2)_B (ordinary rotations x rotations of the extra space x0 x5 x6 x7) as (2; 2, 1) + (2; 1, 2); [A_a, B_b] = 0 to 0e0, residual 6e-16
[PASS] 71-Q2 (numerical) 7+1, two identical particles in an s wave (28 states allowed by the Pauli principle): under SO(7) they form 7 + 21 (Casimir 6 and 10); there is NO SO(7) singlet. Under SO(3) x SU(2)_A x SU(2)_B: 7 = (1; 2, 2) + (3; 1, 1), 21 = (1; 3, 1) + (1; 1, 3) + (3; 1, 1) + (3; 2, 2). The pairs of ordinary spin 0 are the 10 states (1; 2, 2) + (1; 3, 1) + (1; 1, 3) (residual 3e-13)
[PASS] 71-Q3 (exact) 7+1: for space indices i != j, gamma1671[i].gamma1671[j] = diag(tau1671^i tau1671^j, tau1671^i tau1671^j) in 8x8 blocks, so on the particles at rest, u = (e; e)/sqrt(2), the rotation generator Sigma1671^(ij) acts as the real antisymmetric 8x8 matrix (1/2) tau1671^i tau1671^j
[PASS] 71-Q4 (exact) 7+1: the 7 matrices tau1671^j and the 21 matrices tau1671^i tau1671^j (i < j; space indices) are antisymmetric and linearly independent: together they span all 28 antisymmetric 8x8 matrices, i.e. all s-wave pair amplitudes
[PASS] 71-Q5 (exact) 7+1: the quadratic Casimir operator of SO(7), Cas7(Phi) := -sum_(i<j) [Sigma1671^(ij), [Sigma1671^(ij), Phi]], gives Cas7(Phi) = 6 Phi for every Phi = tau1671^j (the vector, 7) and Cas7(Phi) = 10 Phi for every Phi = tau1671^i tau1671^j (the adjoint, 21)
[PASS] 71-Q6 (exact) 7+1: the linear system [tau1671^i tau1671^j, X] = 0 (all 21 planes) for an 8x8 matrix X has rank 63 and is solved by X = 1: the only rotation-invariant pair matrix is the unit matrix, which is SYMMETRIC and therefore excluded by the Pauli principle. Two identical particles of the field cannot form an s-wave pair that is a singlet under all rotations of the seven space dimensions
[PASS] 71-Q7 (exact) 7+1: the ten antisymmetric matrices tau1671^0, tau1671^5, tau1671^6, tau1671^7 and tau1671^a tau1671^b (a < b in {0, 5, 6, 7}) are linearly independent and commute with the generators tau1671^2 tau1671^3, tau1671^3 tau1671^1, tau1671^1 tau1671^2 of ordinary rotations; the system [tau1671^i tau1671^j, Phi] = 0 (ordinary rotations) on antisymmetric Phi has rank 18 = 28 - 10: these ten matrices are ALL the s-wave pairs of ordinary spin 0
[PASS] 71-Q8 (exact) 7+1: each of tau1671^j (any space index j) and tau1671^i tau1671^j (i != j) is an orthogonal matrix: Phi.transpose(Phi) = 1. A condensate whose pair matrix is proportional to tau1671^j or to tau1671^i tau1671^j therefore gives the SAME gap to all eight spin states (a 'unitary' state)
[PASS] 71-Q9 (exact) 7+1: two-particle content of the field bilinears, u_s^T X u_r for particles at rest: X = Cm1671.gamma1671[j] gives -tau1671^j (space index j) and X = Cm1671.gamma1671[4] gives 0; X = Cp1671.gamma1671[i].gamma1671[j] gives tau1671^i tau1671^j. Hence the vector pair field is DeltaV1671^j := dirac1671^T Cm1671 gamma1671^j dirac1671 (its time component contains no pair of two particles) and the adjoint pair field is DeltaT1671^(ij) := dirac1671^T Cp1671 gamma1671^i gamma1671^j dirac1671
[PASS] 71-Q10 (exact) 7+1: the system [tau1671^i tau1671^j, Phi] = 0 for the six rotation planes (1,2), (1,3), (2,3) of ordinary space and (5,6), (5,7), (6,7) of the extra space, on antisymmetric Phi, has rank 27 = 28 - 1, and Phi = tau1671^0 solves it: EXACTLY ONE s-wave pair is a singlet both of the ordinary rotations and of the rotations of x5 x6 x7. Its pair matrix is tau1671^0, the two-particle content of DeltaV1671^0 = dirac1671^T Cm1671 gamma1671^0 dirac1671, the component of the vector pair field along the hidden space dimension x0 (the same result as in 4+4, check 44-Q4)
[PASS] 44-Q1 (numerical) 4+4, fields independent of x5 x6 x7, one particle at rest (gamma1644[4] u = i u, 8 states; deviation 0e0): under SU(2)_X x SU(2)_Y (the two factors of the rotation group SO(4) of the four space coordinates x0 x1 x2 x3; they are NOT the factors SU(2)_A x SU(2)_B of the internal symmetry of the zero-mode sector, which do not act on ordinary space) x SU(2)_T (rotations of the extra times) the states are (2, 1; 2) with one sign of W1644 and (1, 2; 2) with the other; every state has ordinary spin 1/2 (residual 2e-16)
[PASS] 44-Q2 (numerical) 4+4, fields independent of x5 x6 x7, two identical particles in an s wave (28 states): (1, 1; 3) twice, (3, 1; 1), (1, 3; 1), (2, 2; 1), (2, 2; 3) under SU(2)_X x SU(2)_Y x SU(2)_T (X, Y as in 44-Q1). The pairs that are singlets under ALL rotations of the four space dimensions are the 6 states (1, 1; 3) + (1, 1; 3): two triplets of the extra-time rotations, none of them a singlet of SU(2)_T. 10 states have ordinary spin 0 (residual 1e-14)

```
[PASS] 44-Q3  (numerical) 4+4: among the 120 Pauli-allowed bilinears dirac1644^T Cm1644
 ↪  gamma1644[A1]...gamma1644[Ar] dirac1644 (A1 < ... < Ar), those whose two-particle content
 ↪  (particles at rest) lies entirely in the SO(4)-singlet pairs have the index sets {A1,...,Ar} =
 ↪  {5} {0,1,2,3,5} {4,5} {0,1,2,3,4,5} {6} {0,1,2,3,6} {4,6} {0,1,2,3,4,6} {7} {0,1,2,3,7} {4,7}
 ↪  {0,1,2,3,4,7}; they include the components A = 5, 6, 7 (the extra-time components) of the
 ↪  vector pair field DeltaV1644^A := dirac1644^T Cm1644 gamma1644^A dirac1644
[PASS] 44-Q4  (numerical) 4+4: exactly 1 of the 28 pair states is a singlet both of the ordinary
 ↪  rotations (x1 x2 x3) and of the extra-time rotations (x5 x6 x7); it belongs to the space
 ↪  vector (2, 2; 1), is made of one particle with W1644 = +1 and one with W1644 = -1, and is the
 ↪  two-particle content of the bilinears with the index sets {A1,...,Ar} = {0} {0,4}: in
 ↪  particular of DeltaV1644^0 = dirac1644^T Cm1644 gamma1644^0 dirac1644, the component of the
 ↪  vector pair field along the hidden space dimension x0
[PASS] ALL  (exact) every check above passed
```

A.5 The independent verification

The file `data/deflating_checks_independent.txt`, written by `code/scripts/verify_deflating_independent.py`. The checks `I..-G..`, `I..-C..` and `I..-D1` are symbolic identities in the rates, the constants, the scale factors and the wavenumbers; the others are numerical.

```
PROJECT spacetimes, Revision_Deflation: independent verification (computer algebra and
 ↪  general-purpose numerical tools)
generated by: code/scripts/verify_deflating_independent.py (sympy 1.14.0, scipy, numpy 2.5.3,
 ↪  mpmath 1.3.0)
43 of 43 checks passed

[PASS] I44d-G1  4+4, arbitrary rates and constants (computer algebra): the Christoffel symbols
 ↪  obtained from the metric by differentiation are Gamma^t_(4t) = -rate_t and Gamma^4_(tt) =
 ↪  eta_44 eta_tt rate_t b_t^2; all others vanish
[PASS] I44d-G2  4+4: Ricci tensor R^4_4 = -eta_44 gsq, R^t_t = -eta_44 rate_t gsum, zero along
 ↪  x0..x3; curvature scalar -eta_44 (gsq + gsum^2); Einstein tensor eta_44 gprod (component 44),
 ↪  eta_44 [(gsq + gsum^2)/2 - rate_t gsum] (components tt), eta_44 (gsq + gsum^2)/2 (x0..x3):
 ↪  identities in the six symbols g55, g66, g77, const5, const6, const7
[PASS] I44d-G3  4+4: Riemann tensor in the orthonormal frame, R^t_(4t4) = -rate_t^2 and R^t_(sts)
 ↪  = -eta_44 eta_ss rate_t rate_s; no component carries an index 0, 1, 2 or 3
[PASS] I44d-G4  4+4: the Lie derivative of the metric vanishes identically along 14 vector fields
 ↪  (7 translations, 6 rotations of x0..x3, the scaling d_4 + sum rate_t xt d_t); it does not
 ↪  vanish for the rotation of x0 and x5 nor for the boost of x4 and x1; for g55 = g66 = g77 it
 ↪  vanishes in addition along the three rotations of x5, x6, x7 and three special vector fields
 ↪  (20 in all)
[PASS] I44d-G5  4+4: the list of Killing vectors is complete. The matrices of the Lorentz algebra
 ↪  of the tangent space that leave the Riemann tensor unchanged form a space whose dimension,
 ↪  plus 8, bounds the number of Killing vectors from above; exact ranks for (g55, g66, g77) = (1,
 ↪  2, 3): 6 + 8 = 14; (1, 2, 4): 6 + 8 = 14; (3, 5, 7): 6 + 8 = 14; (2, 2, 5): 7 + 8 = 15; (5, 2,
 ↪  2): 7 + 8 = 15; (1, 1, 1): 12 + 8 = 20; (7, 7, 7): 12 + 8 = 20: the bounds 14 (three different
 ↪  rates), 15 (two equal) and 20 (three equal) equal the numbers of Killing vectors exhibited
[PASS] I44d-L1  4+4, Einstein-Gauss-Bonnet gravity (metric with positive space-like components;
 ↪  standard formula of the Lanczos tensor, computer algebra): Lanczos^a_a = -4 g55 g66 g77 gsum
 ↪  for a = 0..3 and Lanczos^4_4 = Lanczos^t_t = 0, identically in the rates and constants; for
 ↪  g55 = g66 = g77 the deflating spacetime solves G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB
 ↪  Lanczos^mu_nu = 0 with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2); for unequal rates it
 ↪  does not, because G^5_5 - G^6_6 = +-(g55 - g66) gsum while the Lanczos tensor vanishes there
[PASS] I44d-C1  4+4: the coordinate matrices gamma^mu = e_A^mu gamma^A satisfy gamma^mu gamma^nu +
 ↪  gamma^nu gamma^mu = 2 g^(mu nu) for all 64 pairs, identically in b_5, b_6, b_7
[PASS] I44d-C2  4+4: the spin connection computed from the tetrad postulate is antisymmetric in
 ↪  its frame indices and equals Omega_t = -(1/2) eta_tt rate_t b_t gamma^t gamma^4 (frame
 ↪  matrices) for t = 5, 6, 7, zero otherwise
[PASS] I44d-C3  4+4: d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for
 ↪  all 64 pairs
[PASS] I44d-C4  4+4: sum_mu gamma^mu Omega_mu = -(gsum/2) gamma^4: the Dirac operator is gamma^4
 ↪  (d_4 - gsum/2) + sum_a gamma^a d_a + sum_t gamma^t d_t / b_t
```

```
[PASS] I44d-C5  4+4: (1/2) gamma^mu gamma^nu (d_mu Omega_nu - d_nu Omega_mu + [Omega_mu,
↪  Omega_nu]) = -(R/4) 1 with R = -eta_44 (gsq + gsum^2)
[PASS] I44d-D1  4+4: the generator K = -m gamma^4 + i k_a gamma^4 gamma^a + i (k_t/b_t) gamma^4
↪  gamma^t of the evolution of the weighted field satisfies K^2 = (k_d^2 - E_perp^2) 1
↪  identically in the eight symbols m, k_0..k_3, k_5, k_6, k_7 and in b_5, b_6, b_7
[PASS] I71d-G1  7+1, arbitrary rates and constants (computer algebra): the Christoffel symbols
↪  obtained from the metric by differentiation are Gamma^t_(4t) = -rate_t and Gamma^4_(tt) =
↪  eta_44 eta_tt rate_t b_t^2; all others vanish
[PASS] I71d-G2  7+1: Ricci tensor R^4_4 = -eta_44 gsq, R^t_t = -eta_44 rate_t gsum, zero along
↪  x0..x3; curvature scalar -eta_44 (gsq + gsum^2); Einstein tensor eta_44 gprod (component 44),
↪  eta_44 [(gsq + gsum^2)/2 - rate_t gsum] (components tt), eta_44 (gsq + gsum^2)/2 (x0..x3):
↪  identities in the six symbols g55, g66, g77, const5, const6, const7
[PASS] I71d-G3  7+1: Riemann tensor in the orthonormal frame, R^t_(4t4) = -rate_t^2 and R^t_(sts)
↪  = -eta_44 eta_ss rate_t rate_s; no component carries an index 0, 1, 2 or 3
[PASS] I71d-G4  7+1: the Lie derivative of the metric vanishes identically along 14 vector fields
↪  (7 translations, 6 rotations of x0..x3, the scaling d_4 + sum rate_t xt d_t); it does not
↪  vanish for the rotation of x0 and x5 nor for the boost of x4 and x1; for g55 = g66 = g77 it
↪  vanishes in addition along the three rotations of x5, x6, x7 and three special vector fields
↪  (20 in all)
[PASS] I71d-G5  7+1: the list of Killing vectors is complete. The matrices of the Lorentz algebra
↪  of the tangent space that leave the Riemann tensor unchanged form a space whose dimension,
↪  plus 8, bounds the number of Killing vectors from above; exact ranks for (g55, g66, g77) = (1,
↪  2, 3): 6 + 8 = 14; (1, 2, 4): 6 + 8 = 14; (3, 5, 7): 6 + 8 = 14; (2, 2, 5): 7 + 8 = 15; (5, 2,
↪  2): 7 + 8 = 15; (1, 1, 1): 12 + 8 = 20; (7, 7, 7): 12 + 8 = 20: the bounds 14 (three different
↪  rates), 15 (two equal) and 20 (three equal) equal the numbers of Killing vectors exhibited
[PASS] I71d-L1  7+1, Einstein-Gauss-Bonnet gravity (metric with positive space-like components;
↪  standard formula of the Lanczos tensor, computer algebra): Lanczos^a_a = -4 g55 g66 g77 gsum
↪  for a = 0..3 and Lanczos^4_4 = Lanczos^t_t = 0, identically in the rates and constants; for
↪  g55 = g66 = g77 the deflating spacetime solves G^mu_nu + Lambda_8 delta^mu_nu + alpha_GB
↪  Lanczos^mu_nu = 0 with Lambda_8 = 3 g55^2 and alpha_GB = -1/(4 g55^2); for unequal rates it
↪  does not, because G^5_5 - G^6_6 = +-(g55 - g66) gsum while the Lanczos tensor vanishes there
[PASS] I71d-C1  7+1: the coordinate matrices gamma^mu = e_A^mu gamma^A satisfy gamma^mu gamma^nu +
↪  gamma^nu gamma^mu = 2 g^(mu nu) for all 64 pairs, identically in b_5, b_6, b_7
[PASS] I71d-C2  7+1: the spin connection computed from the tetrad postulate is antisymmetric in
↪  its frame indices and equals Omega_t = -(1/2) eta_tt rate_t b_t gamma^t gamma^4 (frame
↪  matrices) for t = 5, 6, 7, zero otherwise
[PASS] I71d-C3  7+1: d_mu gamma^nu + Gamma^nu_(mu lam) gamma^lam + [Omega_mu, gamma^nu] = 0 for
↪  all 64 pairs
[PASS] I71d-C4  7+1: sum_mu gamma^mu Omega_mu = -(gsum/2) gamma^4: the Dirac operator is gamma^4
↪  (d_4 - gsum/2) + sum_a gamma^a d_a + sum_t gamma^t d_t / b_t
[PASS] I71d-C5  7+1: (1/2) gamma^mu gamma^nu (d_mu Omega_nu - d_nu Omega_mu + [Omega_mu,
↪  Omega_nu]) = -(R/4) 1 with R = -eta_44 (gsq + gsum^2)
[PASS] I71d-D1  7+1: the Hamiltonian h = m beta + p_j alpha^j + (k_t/b_t) alpha^t of the weighted
↪  field is symmetric and satisfies h^2 = (E_perp^2 + k_d^2) 1 identically in the eight symbols
↪  m, p_0..p_3, k_5, k_6, k_7 and in b_5, b_6, b_7
[PASS] I71d-M1  7+1: psihat = sqrt(u_d) [J_(1/2 + i E_perp/rate_t)(u_d) w + i J_(-1/2 + i
↪  E_perp/rate_t)(u_d) alpha^d w], with h_perp w = E_perp w and u_d = k_d/rate_t, satisfies i d
↪  psihat/dx4 = h psihat (Bessel functions of mpmath; relative residual 1.1e-15 at four values of
↪  x4; mode m = 1, p = (0, 0.3, -0.4, 0.5), k_5 = 0.7, g55 = 2)
[PASS] I71d-M2  7+1: the number of particles created in a mode that starts empty, integrated with
↪  the Runge-Kutta method DOP853 of scipy and projected on the negative-energy states at k_d = 80
↪  E_perp, equals 1/(exp(2 pi E_perp/rate_t) + 1) for all eight internal states in three modes
↪  (largest relative deviation 2.2e-05)
[PASS] I71d-M3  7+1, equal rates 1.3 and a mode that depends on all three deflating coordinates
↪  (constants 1, 0.7, 2): the same law holds (relative deviation 1.4e-05)
[PASS] I44d-M1  4+4: psihat = sqrt(u_d) [K_(1/2 - i E_perp/rate_t)(u_d) w - K_(1/2 + i
↪  E_perp/rate_t)(u_d) i gamma^4 gamma^d w], with K_perp w = i E_perp w and u_d = k_d/rate_t,
↪  satisfies d psihat/dx4 = K psihat (modified Bessel functions of mpmath; relative residual
↪  2.1e-15); the eigenspace of K_perp with the eigenvalue i E_perp has dimension 8, so that there
↪  are eight such solutions; they fall as exp(-u_d)
```

```
[PASS] I44d-M2  4+4: between early x4 and the value at which the WKB exponent is S_wkb = 7.63 the
 ↪  bounded solution falls by exp(-S_wkb) times a constant: ln(amplitude early/late) - S_wkb =
 ↪  0.33534, the exact value (1/2) ln 2 - (1/2) ln(1 + exp(-2 pi E_perp/rate_t)) = 0.33654 (the
 ↪  difference, 1.2e-03, is the correction of relative order 1/u_d to the WKB form)
[PASS] Idef-B1  7+1: a scalar mode that depends on one deflating coordinate and starts empty ends
 ↪  with 1/(exp(2 pi varpi_b/rate_t) - 1) quanta (DOP853 of scipy, three modes, largest relative
 ↪  deviation 4.6e-05)
[PASS] I44d-V1  4+4: the two zero-energy equations of a vortex line of unit winding have a
 ↪  4-dimensional space of constant spinor solutions (singular values, numpy); on it alpha^3 has
 ↪  the eigenvalues -1 -1 +1 +1 and equals the central matrix P_c = i alpha^1 alpha^2 alpha^3 chi:
 ↪  four zero modes, two for each direction along the line
[PASS] I71d-V1  7+1: the two zero-energy equations of a vortex line of unit winding have a
 ↪  4-dimensional space of constant spinor solutions (singular values, numpy); on it alpha^3 has
 ↪  the eigenvalues -1 -1 +1 +1 and equals the central matrix P_c = i alpha^1 alpha^2 alpha^3 chi:
 ↪  four zero modes, two for each direction along the line
[PASS] Idef-C1  cosmology: an independent integration (scipy) of the expansion with three equal
 ↪  rates gives, at fixed 100 theta_* = 1.03958 and with last scattering held at z_* = 1089.92,
 ↪  the Hubble constant for gsum = 0.01 H_0: 66.743 km/s/Mpc (Rust program: 66.743); for gsum =
 ↪  0.1 H_0: 60.773 km/s/Mpc (Rust program: 60.773)
[PASS] Idef-C2  recombination on the deflating background, independently: Peebles' three-level
 ↪  atom (scipy; not the recombination program of the main project) gives z_* = 1090.84 without
 ↪  deflation (Rust program: 1089.92) and 100 theta_* = 1.03905 (1.03965). For gsum = 0.03 H_0:
 ↪  last scattering moves by -0.551 (Rust: -0.550), the helium fraction by -0.00243 (-0.00243);
 ↪  Hubble constant 65.365 km/s/Mpc (Rust: 65.365), and 65.483 (65.483) if last scattering and
 ↪  helium are held fixed; for gsum = 0.1 H_0: last scattering moves by -1.882 (Rust: -1.877), the
 ↪  helium fraction by -0.00844 (-0.00844); Hubble constant 60.367 km/s/Mpc (Rust: 60.368), and
 ↪  60.773 (60.773) if last scattering and helium are held fixed. The shifts of last scattering
 ↪  agree to 0.3 per cent, the changes of the Hubble constant at the fixed angle to 0.01 per cent
 ↪  (full) and 0.00 per cent (last scattering and helium held fixed): the larger fall of the
 ↪  Hubble constant in the full calculation is confirmed by a calculation that shares no
 ↪  recombination code with it
[PASS] I44d-P1  4+4: Einstein tensor of ordinary space to first order in a perturbation (plane
 ↪  wave along x3, synchronous gauge, conformal time, scale factors of x5, x6, x7 unperturbed;
 ↪  arbitrary rates; sympy). With Hc = a'/a, He = Hc - a gsum/2 and the signs chosen so that the
 ↪  time-time component is -(8 pi G) rho: background 3 Hc^2 - 3 a gsum Hc + a^2 gprod (time-time)
 ↪  and -(2 Hc' + Hc^2 - 2 a gsum Hc + a^2 (gsq + gprod)) (space-space); G^4_4: (2 k^2 eta - He
 ↪  h') cos(k x3); G^4_3: 2 k eta' sin(k x3); trace of the space-space part: (2 k^2 eta - 2 He h'
 ↪  - h'') cos(k x3); G^3_3 - G^1_1: (1/2)[(h'' + 6 eta'') + 2 He (h' + 6 eta') - 2 k^2 eta] cos(k
 ↪  x3); all divided by a^2. These are the equations (E), (M), (T), (S) of assumption A3: those of
 ↪  Ma and Bertschinger with Hc replaced by He in three of them. The components G^0_0 and G^5_5
 ↪  along the extra coordinates are perturbed as well: an agent must hold those coordinates
[PASS] I71d-P1  7+1: Einstein tensor of ordinary space to first order in a perturbation (plane
 ↪  wave along x3, synchronous gauge, conformal time, scale factors of x5, x6, x7 unperturbed;
 ↪  arbitrary rates; sympy). With Hc = a'/a, He = Hc - a gsum/2 and the signs chosen so that the
 ↪  time-time component is -(8 pi G) rho: background 3 Hc^2 - 3 a gsum Hc + a^2 gprod (time-time)
 ↪  and -(2 Hc' + Hc^2 - 2 a gsum Hc + a^2 (gsq + gprod)) (space-space); G^4_4: (2 k^2 eta - He
 ↪  h') cos(k x3); G^4_3: 2 k eta' sin(k x3); trace of the space-space part: (2 k^2 eta - 2 He h'
 ↪  - h'') cos(k x3); G^3_3 - G^1_1: (1/2)[(h'' + 6 eta'') + 2 He (h' + 6 eta') - 2 k^2 eta] cos(k
 ↪  x3); all divided by a^2. These are the equations (E), (M), (T), (S) of assumption A3: those of
 ↪  Ma and Bertschinger with Hc replaced by He in three of them. The components G^0_0 and G^5_5
 ↪  along the extra coordinates are perturbed as well: an agent must hold those coordinates
[PASS] Idef-P2  background on the deflating spacetime (sympy; four species: pressureless matter,
 ↪  radiation, a fluid with a variable equation of state, cosmological term): the expression of
 ↪  the program for the derivative of the conformal Hubble rate is the derivative of the equation
 ↪  of the expansion Hc^2 - a gsum Hc + a^2 gprod/3 = (sum of 8 pi G_N a^2 rho)/3 with (8 pi G_N
 ↪  a^2 rho)' = (a gsum - (1 + 3 w) Hc) 8 pi G_N a^2 rho; the quantity B = He' - He^2 + 4 pi G_N
 ↪  a^2 (rho + P) equals a gsum [ (1/6) sum of 8 pi G_N a^2 rho (1 - 3 w) + (a gsum)^2/4 - a^2
 ↪  gprod/3 ] / (2 He), which vanishes without deflation; and the expression of the program for B'
 ↪  is its derivative
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[PASS] Idef-P3  the two identities between the gaps (sympy; pressureless matter, radiation with
↪  anisotropic stress, a fluid with variable equation of state and its own sound speed,
↪  cosmological term; arbitrary rates). With D, T, DP, Pi the sums of 8 pi G_N a^2 times the
↪  density, momentum, pressure and shear perturbations, eta eliminated by the energy constraint
↪  (E), and the equations of motion of matter of Ma and Bertschinger: momentum gap = 2 k^2 eta' -
↪  T, trace gap = h'' + 2 He h' - 2 k^2 eta + 3 DP, shear gap = alpha' + 2 He alpha - eta + (3/2)
↪  Pi/k^2 with alpha = (h' + 6 eta')/(2 k^2) obey  momentum gap = He (trace gap) + h' B - (a
↪  gsum/2)(D - 3 DP)  and  2 k^2 (shear gap) = (trace gap) + (3/k^2)[(momentum gap)' + 2 He
↪  (momentum gap)].  Without deflation B = 0 and a gsum = 0: the three gaps then vanish together,
↪  as in general relativity; with deflation they cannot all vanish
[PASS] Idef-P4  growth of pressureless matter (sympy). From (E) with the momentum constraint:
↪  delta'' + [(2 He' + g)/(2 He)] delta' = [(Hc - a gsum)/(2 He)] g delta (closure "Momentum");
↪  from (E) with the trace equation: delta'' + He delta' = g delta/2, that is, in the time x4,
↪  delta'' + (2 H - gsum/2) delta' = 4 pi G_N rho_m delta (closure "Pressure"); g = 8 pi G_N a^2
↪  rho_m, primes in conformal time. Both hold for every wavenumber. For matter alone and three
↪  equal rates, to the first order in r = gsum/H the growth rate is d ln delta/d ln a = 1 + c r
↪  with c = -9/16 ("Pressure"), -13/16 ("Momentum"), and -11/16 for the four-dimensional equation
↪  delta'' + 2 H delta' = 4 pi G_N rho_m delta, which is neither
[PASS] Idef-P5  growth of pressureless matter from a = 0.01 to today at a fixed Hubble constant,
↪  relative to no deflation: an independent integration (scipy) of the three growth equations
↪  agrees with the Rust program (data/growth_deflation.csv) to 6.3e-07 for the rates up to 0.1
↪  H_0; at gsum = 0.1 H_0: closure "Momentum" 0.95525 (Rust 0.95525); closure "Pressure" 0.97830
↪  (Rust 0.97830); four-dimensional equation 0.96740 (Rust 0.96740)
[PASS] Idef-P6  the three closures as systems of equations for the density contrast of
↪  pressureless matter and the metric (matter and cosmological term; implicit integrator of
↪  scipy; gsum = 0.1 H_0, Hubble constant fixed; growth from a = 0.01 to today relative to no
↪  deflation). Closure "Momentum": 0.955197; closure "Pressure": 0.978268; neither depends on
↪  the wavenumber (to 7e-14 for k = 0.1, 0.03, 0.01, 0.003/Mpc). Closure "Shear" (energy
↪  constraint and equation of the shear; its equations are stiff, because the momentum density of
↪  the agent relaxes at the rate k^2/(3 Hc)): 0.978266 at k = 0.1/Mpc, 0.978251 at k = 0.03/Mpc,
↪  0.978119 at k = 0.01/Mpc, 0.976788 at k = 0.003/Mpc; it differs from the closure "Pressure" by
↪  1.5e-06, 1.7e-05, 1.5e-04, 1.5e-03, that is by 0.31, 0.30, 0.30, 0.27 times (H_0/k)^2: far
↪  inside the horizon the closure "Shear" is the closure "Pressure"
[PASS] Idef-P7  the Einstein-Boltzmann program of the revision against the growth equations: for
↪  the five rates of data/cmb_deflation_growth.csv and both closures, the growth of the density
↪  contrast of matter at k = 0.1/Mpc from a = 0.01 to today in the full calculation (photons,
↪  neutrinos, baryons, cold dark matter), relative to no deflation and at a fixed Hubble
↪  constant, agrees with the independent integration (scipy) of the growth equation of the
↪  closure to 1.3e-04
[PASS] Idef-P8  the agent in the unperturbed universe (sympy; arbitrary rates): if the time-time
↪  Einstein equation holds with matter alone (assumption A2: the agent that keeps the rates
↪  constant has no energy density), the space-space equation and the conservation of matter
↪  require the agent to have the pressure p_agent in ordinary space with 8 pi G_N p_agent =
↪  -(gsum/H) dH/dx4 - 2 gsum H + 2 gprod - gsum gprod/(3 H). To the first order in the rates this
↪  is -gsum H/2 in an era of matter: a tension, of the relative size gsum/H
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References

1. D. Chirkov, S. Pavluchenko and A. Toporensky, *Exact exponential solutions in Einstein–Gauss–Bonnet flat anisotropic cosmology,* Mod. Phys. Lett. A **29**, 1450093 (2014); arXiv:1401.2962.

Appendix B. Files

file	content	written by
`code/deflation-rs/src/metric.rs, geometry.rs`	metric, Christoffel symbols, curvature, Killing vectors, geodesics	—

file	content	written by
code/deflation-rs/src/curved.rs	coordinate matrices, spin connection, Dirac operators and their checks	—
code/deflation-rs/src/modes.rs	integration of the mode equations, creation and extinction	—
code/deflation-rs/src/symmetry_defl.rs, cmd_symmetry.rs	surviving rotations and boosts, internal symmetry	—
code/deflation-rs/src/rat.rs, qmat.rs	exact rational numbers and exact complex-rational matrices	—
code/scripts/verify_deflating_independent.py	the independent verification	—
data/deflating_checks.txt, .csv	the checks of Appendix A.1	deflation matrices
data/flat_matrix_checks.txt, .csv	the checks of Appendix A.4	deflation matrices
data/curvature.csv	Ricci tensor, curvature scalar and Einstein tensor for the five parameter sets	deflation matrices
data/geodesic_1671d.csv, data/geodesic_1644d.csv	the two geodesics of Sec. 1.7	deflation matrices
data/mode_checks.txt, .csv	the checks of Appendix A.2	deflation modes
data/creation_single.csv	created number against $E_\perp/\mathrm{g55}$, one direction	deflation modes
data/creation_anisotropic.csv	created numbers for two directions with different rates	deflation modes
data/creation_history.csv	the number of particles in a mode against time	deflation modes
data/extinction_checks.csv, data/extinction_profile.csv	the five cases of bounded 4+4 solutions; a profile	deflation modes
data/symmetry_checks.txt, .csv, data/isometries.csv	the checks of Appendix A.3; the 28 rotations and boosts	deflation symmetry
data/deflating_checks_independent.txt	the checks of Appendix A.5	the Python script
figures/fig_isometries, fig_geodesics, fig_creation, fig_extinction (.pdf, .png)	the four figures	make_figures.py